Advanced Empirical Research Methods: Empirical IO

Notes for lectures

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Outline: Demand and Supply Modelling I

1. The canonical (general) model.
2. Inversion of market share function.
3. GMM estimation with market-level data.
4. The logit and nested logit models. Applications: beer and autos. Classwork 1,2 & 3: Simulating market data. Inversion. Estimation. Instrumental variables. Post-estimation calculation of elasticities, markups, marginal costs.
5. The mixed-logit (BLP) model. The BLP contraction mapping. Applications: autos and cereals. Classwork 4 and 5.
6. Counterfactual simulation of equilibrium market outcomes. Classwork 6.
7. (Supplementary section) Incorporating a supply side into estimation with the multi-product Nash pricing assumption.
8. (Supplementary section) Adding consumer-level data.
9. (Supplementary section) Nonparametric identification.

References I

- Berry and Haile (2021) FOUNDATIONS OF DEMAND ESTIMATION NBER Working Paper 29305 link
- Berry, S. (1994), Estimating discrete choice model of product differentiation, *Rand Journal of Economics*, 25, 242-263.
- Berry, S., J. Levinsohn and A. Pakes (1995), Automobile Prices in Market Equilibrium, *Econometrica*, 1995, 63, 841–890 [BLP].
- Berry, S., J. Levinsohn and A. Pakes (2004). Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market. *Journal of Political Economy* 112.1
- Brunner, Hess, Romahn, Weiser (2023) Reliable Estimation of Random Coefficient Logit Demand Models forthcoming *Journal of Econometrics* link
- *Conlon, Christopher, and Gortmaker (2020). Best practices for differentiated products demand estimation with PyBLP. *RAND Journal of Economics*, 51 (4), 1108-1161.

References II

- *Gandhi A. and A. Nevo (2021) Empirical Models of Demand and Supply in Differentiated Products Industries NBER Working Paper No. 29257
- Gandhi A. and Houde J-F (2020), MEASURING SUBSTITUTION PATTERNS IN DIFFERENTIATED-PRODUCTS INDUSTRIES, NBER Working Paper 26375,link
- Nevo (2001) Measuring Market Power in the Ready-to-Eat Cereal Industry *Econometrica*, Vol. 69, No. 2.
- Nevo (2004) A Practitioner's Guide to Estimation of Random-Coefficients Logit Models of Demand, *Journal of Economics & Management Strategy* Volume 9, Issue 4
- Reynaert, Verboven, (2014) Improving the performance of random coefficients demand models: The role of optimal instruments, *Journal of Econometrics*, 179:1, 2014, 83-98
- PyBLP give a good overview in their summary page

References III

- Textbooks:
  - Hortacsu and Joo (2024) *Structural Econometric Modelling in Industrial Organization and Quantitative Marketing* Ch 2.2, 3.2.
  - Train, K. (2009) *Discrete Choice Methods with Simulation* Ch 3-6,10,13 on logit, mixed logit, simulation, endogeneity;
  - Cameron and Trivedi (2006) *Miroeconometrics* Ch 6 on linear and nonlinear GMM.

BLPestimatoR for Demand Estimation

- BLPestimatoR is an R implementation of routines for estimating the demand for differentiated products with BLP-type random coefficients logit models. This package was created by Daniel Brunner, Constantin Weiser and Andre Romahn
- For code and further information see https://CRAN.R-project.org/package=BLPestimatoR
- We use this package for BLP modelling in the classwork.
- Students will find it very helpful to read and play around with the code found on the BLPestimator vignette page

PyBLP for Demand Estimation

- PyBLP is a Python 3 implementation of routines for estimating the demand for differentiated products with BLP-type random coefficients logit models. This package was created by Jeff Gortmaker in collaboration with Chris Conlon.
- For code and further information see https://pyblp.readthedocs.io/en/stable/
- It's possible to access this from R using the reticulate package: as discussed here.
- Students (if they prefer to) may use Python and PyBLP instead of R and BLPestimatoR in their submitted work.

Motivation I

Estimation of differentiated-product demand is useful for measuring

1. Alternative models of conduct
2. Market power
3. The social value of a new product
4. The effect of a tax on equilibrium prices and consumption
5. Effect of a merger
6. Effects of trade policy, e.g. quota or tariff
7. Analyzing school or hospital choice

among many other questions. Aim of the lectures is to provide the techniques to study these questions using exercises to give students an understanding of how to code the methods on data.

Demand models: dimensionality and flexibility I

- Let market $t = 1, \ldots, T$ have differentiated products $j = 1, ..., J_t$.
- The market is a location and time for which demand is defined and prices are set.
- There is an outside good $j = 0$, to allow consumers to substitute from the products if all prices increase.
- In many applications J_t is large (e.g. over 100 autos).
- Let $\boldsymbol{s}_t$ and $\boldsymbol{p}_t$ be $J_t \times 1$ vectors of quantities and prices.
- Let $\boldsymbol{s}_t = \boldsymbol{s}_t(\boldsymbol{p}_t)$ be the market level demand system.
- We would like the demand system to be flexible.
- $\boldsymbol{s}_t = \boldsymbol{s}_t(\boldsymbol{p}_t)$ is (second-order) flexible "if at any given $\boldsymbol{p}_t$, parameters can be chosen so $\boldsymbol{s}_t$ and the complete set of own and cross price elasticities are capable of arbitrary values subject only to the constraint of theoretical consistency." [Lau (1986) *Handbook of Econometrics* Vol 3]

Demand models: dimensionality and flexibility II

- An example of a second-order flexible demand system is

$$\boldsymbol{s}_t(\boldsymbol{p}_t) = \beta_0 + \beta_1 \boldsymbol{p}_t \tag{1}$$

  where β_0 is a $J \times 1$ vector and β_1 a $J \times J$ matrix of cross-price effects.
- This can be derived from a quadratic utility system. The number of parameters in (1) is $J^2 + J$, i.e. increasing in J^2, a high dimension for large J. Curse of dimensionality.
- The dimension problem in conventional flexible utility systems (AIDS, quadratic utility, etc.) is a consequence of preferences being specified over products j; discrete choice models avoid this dimension problem by specifying preferences to be over the products' characteristics. This comes at a cost of inflexibility, the extent of which varies by specification.

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The Canonical Model I

- $i = 1, \ldots, I$ consumers, $t = 1, \ldots, T$ markets
- $j = 1, \ldots, J_t$ products.
  - $\boldsymbol{x}_j = (x_{j1}, \ldots, x_{jK})$: $K \times 1$ vector of j's observed characteristics .
  - p_{jt}: price of j in market t
- i's utility for product j in market t is

$$u_{ijt} = \sum_{k=1}^{K} (\beta_k + \nu_{ikt}) x_{jk} - (\alpha + \nu_{i0t}) p_{jt} + \xi_{jt} + \varepsilon_{ijt} \qquad (2)$$

- Mean utility:
  - $\boldsymbol{\beta}$ is $1 \times K$ vector of mean marginal tastes of $\boldsymbol{x}_j$.
  - α is scalar mean marginal utility of money.
  - ξ_{jt} is scalar mean utility of j's unobserved characteristics in market t. We call this unobserved utility.
- Consumer heterogeneity (with distributions H and G):
  - $\boldsymbol{\nu_{it}} \sim H(\boldsymbol{\nu}, \boldsymbol{\sigma}_\nu)$, is i's $1 \times (K+1)$ taste effect vector.
  - $\boldsymbol{\varepsilon_{it}} \sim G(\boldsymbol{\varepsilon}, \boldsymbol{\sigma}_\varepsilon)$ is i's $J_t \times 1$ idiosyncratic taste shocks, has positive support on the set of real numbers $\mathbb{R}^{J_t}$.

The Canonical Model II

- i's utility in t for j can be written

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt} \tag{3}$$

where δ_{jt} is a mean utility index for product j in market t:

$$\delta_{jt} = \boldsymbol{\beta}\boldsymbol{x}_j - \alpha p_{jt} + \xi_{jt}$$

and μ_{ijt} is consumer heterogeneity up to ε_{ijt}

$$\mu_{ijt} = \sum_{k=1}^{K} \nu_{ikt} x_{jk} - \nu_{i0t} p_{jt}$$

- For the outside good ($j = 0$) we assume utility is

$$u_{i0t} = \delta_{0t} + \varepsilon_{i0t}.$$

- Let J_t denote either the number of inside goods, or the set of goods (including the outside good), in market t, depending on context.

The Canonical Model III

- Consumer i in market t maximizes utility choosing best product from set J_t (which includes $j = 0$) so that:

$$(i \text{ in } t \text{ chooses } j) \;\Leftrightarrow (u_{ijt} \geq u_{ikt},\ \forall k \in J_t)$$

- Location and scale of utility are not identified in discrete-choice models so two normalizations are allowed:
  1. (location) we fix $\delta_{0t} = 0$ (and also fix the location of ε in G)
  2. (scale) we fix the variance of ε in G.
- Choice probability (market share) for product j in market t is

$$s_{jt} = \int_{\nu} \int_{\varepsilon} 1[u_{ijt} \geq u_{ikt},\ \forall k \in J_t] \mathrm{d}G(\varepsilon) \mathrm{d}H(\nu) \tag{4}$$

or, in terms of prices $\boldsymbol{p}_t$, characteristics $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_J)$, unobserved utility $\boldsymbol{\xi}_t = (\xi_{1t}, \ldots, \xi_{Jt})$, and parameters $\boldsymbol{\theta}$:

$$s_{jt} = s_{jt}(\boldsymbol{p}_t, \boldsymbol{x}, \boldsymbol{\xi}_t, \boldsymbol{\theta}). \tag{5}$$

- For demand in units multiply by number of consumers $q_{jt} = l_t s_{jt}$.

The Canonical Model IV

- Parameters to estimate are $\boldsymbol{\theta} = (\boldsymbol{\beta}, \alpha, \boldsymbol{\sigma}_\nu, \boldsymbol{\sigma}_\varepsilon)$.
- Number of parameters:
  1. $\boldsymbol{\beta}$: K parameters, one for each product characteristics.
  2. α: 1 parameter on price
  3. $\boldsymbol{\sigma}_\nu$: varies by specification, usually $\leq K + 1$ parameters.
  4. $\boldsymbol{\sigma}_\varepsilon$: varies by specification (usually none, by scale/location fixing).
- So, number of parameters is determined by K not J. Helpfully, the number of parameters does not increase in the square of J as in (1). So the dimensionality problem, discussed above, has been overcome. This is because the preferences in the model are specified in terms of product characteristics not in terms of products. (This statement is not quite true: the ε_{ijt} disturbance is a preference over products, however it's usually specified to come from a distribution with no unknown parameters.)

The Canonical Model V

- We have written the canonical model in its general form. We will shortly discuss four alternative specifications of it (simple logit model, nested logit, probit, mixed logit).
- The taste effects $\boldsymbol{\nu}_i \sim H(\boldsymbol{\nu}, \boldsymbol{\sigma}_\nu)$ that allow heterogeneity in preference for product characteristics including price may be draws either from an assumed form such as normal, or from an empirical distribution of demographic variables (such as income), or a combination of both.
- Flexibility: varies with specification. We will assess the flexibility of the model when we look at alternative specifications of it.

Utility theory foundations I

- The canonical utility function may be interpreted as a standard indirect utility conditional on chosen alternative j.
- It can be written as a function of the consumer's budget y_i.
- Let other (non-inside) goods be summarized by the scalar z_i.
- The canonical utility is implied if direct utility is quasi-linear:

$$u_{ij} = \sum_{k=1}^{K} \beta_{ik} x_{jk} + \alpha_{i0} z_i + \xi_j + \varepsilon_{ij} \tag{6}$$

normalizing the $p_z = 1$, the budget constraint if i chooses j is

$$p_j + z \leq y_i$$

so that substituting into direct utility we get

$$u_{ij} = \sum_{k=1}^{K} \beta_{ik} x_{jk} + \alpha_{i0}(y_i - p_j) + \xi_j + \varepsilon_{ij}. \tag{7}$$

Since $\alpha_{i0} y_i$ is common across j it cancels to give (2).

- It is not necessary to assume quasi-linear utility, e.g. BLP(1995) assume u_{ij} is linear in the log of other consumption: $\ln(y_i - p_j)$.

Utility theory foundations II

- Other assumptions: at most one product chosen, only one *unit* of the product chosen, uniform pricing in any t, complete information (no need for search), static choice.
- An implication of the framework is alternatives are substitutes.
- All of these assumptions can be relaxed with varying levels of complication being added; with some relaxations inversion (see next section) becomes impossible. We do not discuss these relaxations.

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Inverting the market share function

- The canonical demand model is a system of J_t equations in each market t:

$$s_{jt} = s_{jt}(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{\xi}_t, \boldsymbol{\theta}) \text{ for } j = 0, ..., J_t. \tag{8}$$

- Since goods are substitutes, $s_{jt}(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{\xi}_t, \boldsymbol{\theta})$ is increasing in ξ_{jt} and decreasing in ξ_{kt} for $k \neq j$.
- Therefore we can think of $\boldsymbol{\xi}_t$ as residuals: they adjust to fit the observed market shares given any parameters.
- We naturally think of market share s_{jt} as the "left hand side" variable and the unit of observation as the individual product-market (j, t).

Rationale for inversion

- The form of (8)

$$s_{jt} = s_{jt}(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{\xi}_t, \boldsymbol{\theta}) \text{ for } j = 0, ..., J_t \tag{9}$$

  is not convenient for estimation using ξ_{jt} as a residual:
  - ξ_{jt} is not additively-separated from the other terms in the equation. This makes it hard to use standard GMM estimation.
  - All the $\boldsymbol{\xi}_t = (\xi_{1t}, \ldots \xi_{Jt})$ appear in the equation for s_{jt}. But to use standard estimation techniques we want one error per observation.
- We would like a form in which
  - the $\boldsymbol{\xi}$ enter additively and
  - there is a single ξ_{jt} per observation.
- This can be achieved by inverting the equation.

Using the utility index

- Vectorize (8) as a system of J_t equations for market t:

$$\boldsymbol{s}_t = \boldsymbol{s}_t(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{\xi}_t, \boldsymbol{\theta}). \tag{10}$$

- Recall the utility index for each (j, t):

$$\delta_{jt} = \beta \boldsymbol{x}_j - \alpha p_{jt} + \xi_{jt}.$$

- Parameters $\boldsymbol{\theta} = (\beta, \alpha, \boldsymbol{\sigma})$ are in two groups: mean utility parameters (β, α) enter δ_{jt} and nonlinear parameters $(\boldsymbol{\sigma})$ dont.
- As $(\beta, \alpha, \boldsymbol{\xi}_t)$ enter (10) exclusively via $\boldsymbol{\delta}$ we can write:

$$\boldsymbol{s}_t = \boldsymbol{s}_t(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{\delta}_t, \boldsymbol{\sigma}) \tag{11}$$

- Under some conditions, market share functions (10) can be inverted to obtain utility index (aka inverse market share) functions:

$$\boldsymbol{\delta}_t = \boldsymbol{\delta}_t(\boldsymbol{x}, \boldsymbol{p}_t, \boldsymbol{s}_t, \boldsymbol{\sigma}) \tag{12}$$

Invertibility conditions

- In this slide we condition on the variables $(\boldsymbol{x}, \boldsymbol{p}_t)$ and the market t so they are suppressed in notation.
- Berry (1994) shows that under the following invertibility conditions there is always a unique $\boldsymbol{\delta}$ for any $\boldsymbol{s}$ and $\boldsymbol{\sigma}$ so we can invert $\boldsymbol{s} = \boldsymbol{s}(\boldsymbol{\delta}, \boldsymbol{\sigma})$ to obtain $\boldsymbol{\delta} = \boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{\sigma})$:

1. $\boldsymbol{s}(\boldsymbol{\delta}, \boldsymbol{\sigma})$ is differentiable in $\boldsymbol{\delta}$.
2. $\boldsymbol{s}(\boldsymbol{\delta}, \boldsymbol{\sigma})$ obeys $\partial s_j / \partial \delta_j > 0$ and $\partial s_j / \partial \delta_k < 0$ for $j \neq k$.
3. $s_j(\delta, \sigma) \to 1$ as $\delta_j \to \infty$ and $s_j(\delta, \sigma) \to 0$ as $\delta_j \to -\infty$.

- These conditions are satisfied by the specific discrete choice models we consider in the course.

Recovering the separated residuals as functions of θ

- For any observed $\boldsymbol{s}_t$ and parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}, \alpha, \boldsymbol{\sigma})$ we now have $\delta_{jt} = \delta_{jt}(\boldsymbol{s}_t, \boldsymbol{\sigma}, \boldsymbol{x}, \boldsymbol{p}_t)$. Substituting into $\delta_{jt} = \boldsymbol{\beta}\boldsymbol{x}_j - \alpha p_{jt} + \xi_{jt}$ and rearranging we get

$$\xi_{jt}(\boldsymbol{\theta}) = \delta_{jt}(\boldsymbol{s}_t, \boldsymbol{\sigma}, \boldsymbol{x}, \boldsymbol{p}_t) - (\boldsymbol{\beta}\boldsymbol{x}_j - \alpha p_{jt}). \tag{13}$$

- We now have a single additively-separate error ξ_{jt} for each unit of observation (j, t) as desired for GMM estimation.
- Parameters $\boldsymbol{\sigma}$ enter non-linearly, the others linearly.
- For some utility specifications (e.g. logit model) all parameters are linear.

Note also: the model has a perfect fit in terms of observed and predicted market shares. The ξ terms adjust to ensure the market share data is fully matched to the model.

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GMM with market level data

- Market level data comprise (for each product and market observation $(j > 0, t)$) the observed market shares, product characteristics and prices:

$$\boldsymbol{w}_{jt} = \{s_{jt}, \boldsymbol{x}_{jt}, p_{jt}, \boldsymbol{z}_{jt}\} \text{ for all } (j > 0, t).$$

- The number of products J_t may vary across markets (unbalanced panel). This is considered helpful for identifying parameters because the substitution patterns are informative.
- Observed s_{jt} constructed using $s_{jt} = q_{jt}/I_t$ where q_{jt} is qty and I_t is potential consumers; specification of I_t not always obvious.
- To simplify notation we initially assume one market and suppress t. Then the number of observations is J where $j = 1, \ldots, J$.
- $\boldsymbol{z}_j$ is a vector of $M \geq dim(\boldsymbol{\theta}) \equiv L$ instrumental variables with individual elements:

$$\{z_{m,j}\} \text{ for all } (j > 0) \text{ and } m = 1, \ldots, M.$$

GMM with market level data I

- Let $\boldsymbol{\theta}_0 = (\beta_0, \alpha_0, \boldsymbol{\sigma}_0)$ be parameters that generate the data.
- Recap: canonical model's inverse market share function (13) is

$$\xi_j(\boldsymbol{\theta}, \boldsymbol{s}, \boldsymbol{x}, \boldsymbol{p}) = \delta_j(\boldsymbol{s}, \boldsymbol{\sigma}, \boldsymbol{x}, \boldsymbol{p}) - (\beta \boldsymbol{x}_j - \alpha p_j). \tag{14}$$

- We assume population moments satisfy the conditional mean-independence moment assumption:

$$E[\xi_j(\boldsymbol{\theta}_0, \boldsymbol{s}, \boldsymbol{x}, \boldsymbol{p}) | \boldsymbol{z}_j] = 0. \tag{15}$$

  where expectation is with respect to the data $\boldsymbol{w}_j$ and $\boldsymbol{z}_j = (z_{1j}, \ldots, z_{Mj})$ is $1 \times M$.
- This implies M (unconditional) zero-covariance conditions:

$$E[z_{mj}\xi_j(\boldsymbol{\theta}_0, \boldsymbol{s}, \boldsymbol{x}, \boldsymbol{p})] = 0 \text{ for } m = 1, \ldots, M. \tag{16}$$

  (This implication is from the Law of Iterated Expectations: $E[\mathbf{z}\xi] = E_{\mathbf{z}}[E[\mathbf{z}\xi|\mathbf{z}]] = E_{\mathbf{z}}[\mathbf{z}E[\xi|\mathbf{z}]] = E_{\mathbf{z}}[\mathbf{z}0] = \mathbf{0}$).

GMM with market level data II

- The sample covariance moments corresponding to (16) are

$$(1/J)\sum_{j=1}^{J}\xi_j(\boldsymbol{\theta},\boldsymbol{s},\boldsymbol{x},\boldsymbol{p})z_{m,j} \text{ for } m=1,\ldots,M. \tag{17}$$

- For each j there is an $M\times 1$ sample vector

$$\boldsymbol{g}(\boldsymbol{\theta},\boldsymbol{w}_j)=\begin{bmatrix}\xi_j(\boldsymbol{\theta},\boldsymbol{s},\boldsymbol{x},\boldsymbol{p})z_{1,j}\\ \vdots\\ \xi_j(\boldsymbol{\theta},\boldsymbol{s},\boldsymbol{x},\boldsymbol{p})z_{M,j}\end{bmatrix}$$

- GMM estimates are given by

$$\hat{\boldsymbol{\theta}}=\arg\min_{\boldsymbol{\theta}}\left[\sum_{j=1}^{J}\boldsymbol{g}(\boldsymbol{\theta},\boldsymbol{w}_j)\right]'\boldsymbol{W}\left[\sum_{j=1}^{J}\boldsymbol{g}(\boldsymbol{\theta},\boldsymbol{w}_j)\right] \tag{18}$$

where the minimand is a weighted sum of the squares and cross-products of the M sample moments, and $\boldsymbol{W}$ is a symmetric positive-definite $M\times M$ weighting matrix. (Positive-definiteness ensures minimand is positive unless moments are all zero).

Standard Errors

- Standard GMM: if
  1. Data w_j are iid across j,
  2. $\boldsymbol{\theta}$ is $L \times 1$ with $L \leq M$,
  3. $E[\boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)] = \mathbf{0}$
  4. $\boldsymbol{G} = E[\frac{\partial}{\partial \boldsymbol{\theta}} \boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)]$, which is $M \times L$, has rank at least L,

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- then,
  - $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ as $J \to \infty$

Standard Errors

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- then,
  - $\hat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$ as $J \to \infty$
  - $\sqrt{J}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(0, \boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{A}^{-1})$ as $J \to \infty$, where

$$\boldsymbol{A} = \boldsymbol{G}'\boldsymbol{W}\boldsymbol{G}, \qquad \boldsymbol{B} = \boldsymbol{G}'\boldsymbol{W}\boldsymbol{V}\boldsymbol{W}\boldsymbol{G}, \qquad \boldsymbol{V} = E[\boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)\boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)'].$$

Standard Errors

- Standard GMM: if
  1. Data w_j are iid across j,
  2. $\boldsymbol{\theta}$ is $L \times 1$ with $L \leq M$,
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$$\boldsymbol{A} = \boldsymbol{G}'\boldsymbol{W}\boldsymbol{G}, \qquad \boldsymbol{B} = \boldsymbol{G}'\boldsymbol{W}\boldsymbol{V}\boldsymbol{W}\boldsymbol{G}, \qquad \boldsymbol{V} = E[\boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)\boldsymbol{g}(\boldsymbol{\theta}_0, \boldsymbol{w}_j)'].$$

- $\widehat{Avar}(\hat{\boldsymbol{\theta}}) = \hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{B}}\hat{\boldsymbol{A}}^{-1}/J$, where

$$\hat{\boldsymbol{A}} = \hat{\boldsymbol{G}}'\boldsymbol{W}\hat{\boldsymbol{G}}, \qquad \hat{\boldsymbol{B}} = \hat{\boldsymbol{G}}'\boldsymbol{W}\hat{\boldsymbol{V}}\boldsymbol{W}\hat{\boldsymbol{G}},$$

$$\hat{\boldsymbol{G}} = \frac{1}{J}\sum_j \frac{\partial}{\partial \hat{\boldsymbol{\theta}}} \boldsymbol{g}(\hat{\boldsymbol{\theta}}, \boldsymbol{w}_j), \qquad \hat{\boldsymbol{V}} = \frac{1}{J}\sum_j \boldsymbol{g}(\hat{\boldsymbol{\theta}}, \boldsymbol{w}_j)\boldsymbol{g}(\hat{\boldsymbol{\theta}}, \boldsymbol{w}_j)'.$$

- Standard errors, s.e.$(\hat{\boldsymbol{\theta}})$, are sq. roots of the diagonal of $\widehat{Avar}(\hat{\boldsymbol{\theta}})$.

Adding markets

- We dropped t to avoid clutter. Adding t, all the previous expressions are exactly the same, except now with jt as the unit of observation, so in subscripts we replace j with jt and we divide by $\sum_t J_t$ observations instead of J (in notation we will assume for simplicity this is JT i.e. a balanced panel).
- Thus the sample covariance moments (17) are now

$$(1/JT)\sum_{t=1}^{T}\sum_{j=1}^{J}\xi_{jt}(\boldsymbol{\theta},\boldsymbol{s}_t,\boldsymbol{x},\boldsymbol{p}_t)z_{m,jt} \text{ for } m=1,\ldots,M. \quad (19)$$

- Sometimes we wish to cluster on product j, which avoids assuming independence between observations for the same j in different t. This impacts only the standard errors: in place of

$$\hat{\boldsymbol{V}} = \tfrac{1}{JT}\sum_j\sum_t \boldsymbol{g}(\hat{\boldsymbol{\theta}},\boldsymbol{w}_{jt})\boldsymbol{g}(\hat{\boldsymbol{\theta}},\boldsymbol{w}_{jt})'. \quad (20)$$

we instead use

$$\hat{\boldsymbol{V}} = \tfrac{1}{J}\sum_j\left[\sum_t \boldsymbol{g}(\hat{\boldsymbol{\theta}},\boldsymbol{w}_{jt})\right]\left[\sum_t \boldsymbol{g}(\hat{\boldsymbol{\theta}},\boldsymbol{w}_{jt})\right]'.$$

Efficiency I

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- The efficiency of the estimator depends on choice of $\boldsymbol{W}$.
- Hansen (1982): a consistent estimator of $\boldsymbol{V}^{-1}$ gives estimates with the lowest variance. A two-step approach is standard, using the 2SLS weighting matrix

$$\boldsymbol{W} = (\boldsymbol{Z}'\boldsymbol{Z})/(JT), \tag{21}$$

in the first step, where $\boldsymbol{Z}$ is the $JT \times M$ matrix of instruments, and in the second step the model is run again using estimates from the first step to calculate

$$\boldsymbol{W} = \hat{\boldsymbol{V}}^{-1}.$$

Efficiency II

- Efficiency also depends on the choice of instruments. Note that

$$E[\xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t, \boldsymbol{x}, \boldsymbol{p}_t)|\boldsymbol{z}_j] = 0 \tag{22}$$

  implies the M conditions

$$E[\mathbf{A}_{jt}(\mathbf{z}_j)\xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t, \boldsymbol{x}, \boldsymbol{p}_t)] = \mathbf{0}. \tag{23}$$

  where $\mathbf{A}_{jt}(\mathbf{z}_j)$ is an M-vector which is a function of $\mathbf{z}_j$. The investigator can construct many possible instrument functions.
- Chamberlain (1987) shows optimal instruments are

$$\mathbf{A}_{jt}(\mathbf{z}_j) = E\left[\frac{\partial \xi_{jt}(\boldsymbol{\theta}_0)}{\partial \theta'}|\boldsymbol{z}_j\right] \qquad (1 \times L).$$

- This is for single-equation GMM i.e. no supply side equation being estimated, see Reynaert and Verboven (2014, sec 2.5).
- Snag: need θ_0. But, might be able to approximate (see later).

Efficiency III

- Finally, the standard errors should also be adjusted to account for simulation error in the moments (see discussion in Train's textbook for details).

Comments on asymptotics I

- Market level data are:

$$\boldsymbol{w}_{jt} = \{s_{jt}, \boldsymbol{x}_{jt}, p_{jt}, \boldsymbol{z}_{jt}\} \text{ for all } j > 0, t.$$

- The estimation uses sample covariance moments of the form $g_m(\boldsymbol{\theta}) = \sum_{t=1}^{T} \sum_{j=1}^{J} \xi_j(\boldsymbol{\theta}) z_{j,m}$ (assuming balanced panel for simple exposition).
- In mixed logit, simulation of market shares $\boldsymbol{s}_t$, and numerical inversion of market shares to get ξ_{jt}, are both required.
- The number of observations is JT and asymptotic distribution of estimators obtained by taking limit as $TJ \to \infty$, or limit as $J \to \infty$ if clustering on product.
- There are two ways this is non-standard
  1. As J increases the existing observations change: market shares $\boldsymbol{s}$ fall and equilibrium prices $\boldsymbol{p}$ change (because of competition). Hence, there is interdependence between observations in the data generating process.

Comments on asymptotics II

2. As J increases, for models where market shares are simulated with some finite number of draws NS, because there is not an analytical form for the integral in (5), the simulation error $(\boldsymbol{s}^{sim} - \boldsymbol{s})$ becomes larger relative to the market share $\boldsymbol{s}$ being simulated.

- These asymptotic issues are discussed in: Berry, Linton, Pakes (2004) "Limit Theorems for Estimating the Parameters of Differentiated Product Demand Systems" *RES* and Ackerberg et al (2007) Econometric Tools for Analyzing Market Outcomes.
- Berry, Linton, and Pakes (2004) show that in a single cross-section the estimator satisfies asymptotic normality if the ratio of the number of products squared over the number of simulation draws, denoted J^2/NS, is bounded as the number of products becomes large.

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Simple Logit Model; Type-1 EV Distribution I

- Suppress t subscripts. The simple logit model does not have any heterogeneity other than ε_{ij} so that

$$u_{ij} = \delta_j + \varepsilon_{ij} \tag{24}$$

- The ε_{ij} in logit is iid Type-1 EV. This has CDFs and PDFs:

$$G(\varepsilon, \gamma) = \exp(-\exp(-(\varepsilon - \gamma))) \tag{25}$$

$$g(\varepsilon, \gamma) = \exp(\gamma - \varepsilon)\exp(-\exp(-\varepsilon - \gamma)) \tag{26}$$

- Support on $\mathbb{R}$. Density shown in Figure 1.
- $E(\varepsilon) = \gamma + 0.57722$ (Euler's constant). Variance $\pi^2/6$.
- ε_{ij} is iid for each (i, j).
- We have normalized the variance parameter since the scale of utility is not identified in discrete choice models.
- By permitted location normalization we set $\gamma = 0$ in choice estimation.
- When $\gamma = 0$ we write $G(\varepsilon, \gamma) = G(\varepsilon)$.

Simple Logit Model; Type-1 EV Distribution II

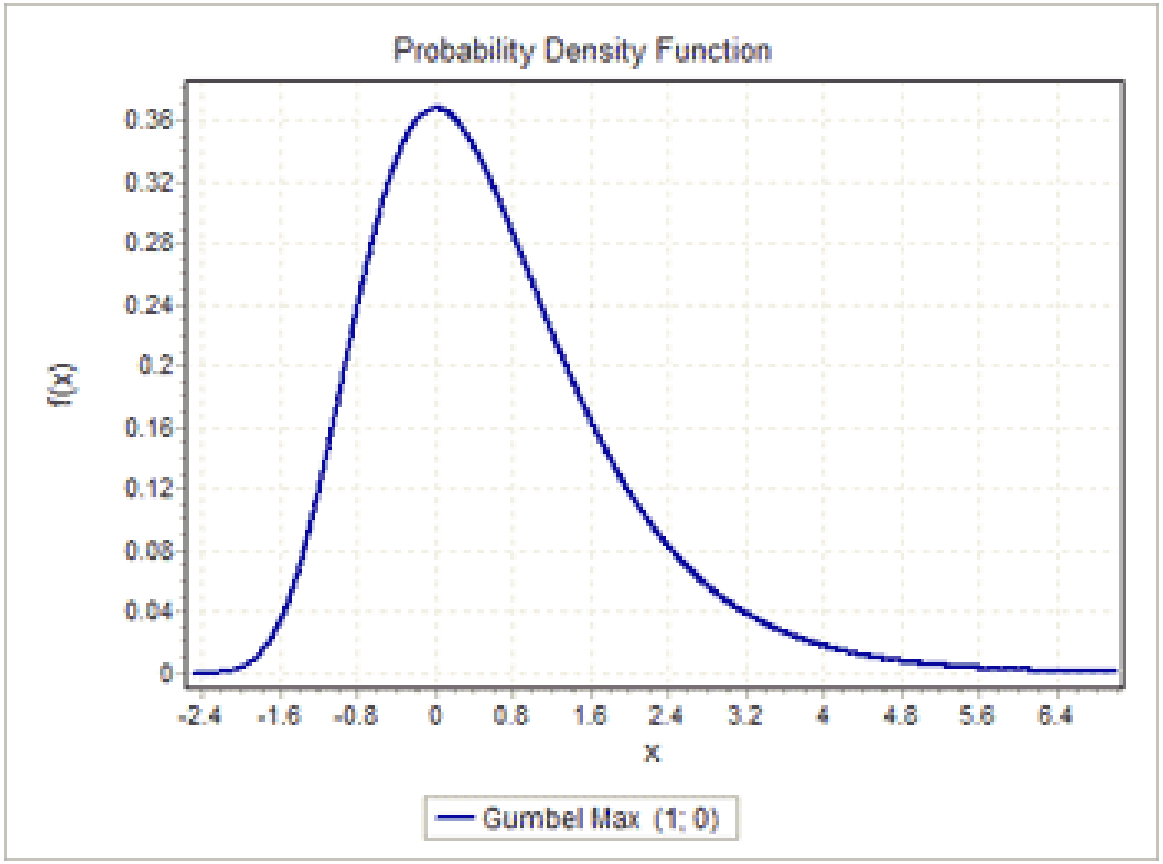

Figure: Type-1 EV Density

Simple logit: consumer surplus I

- Since we fix location parameter of ε_{ij} to zero it follows that $u_{ij} = \delta_j + \varepsilon_{ij}$ is T1EV with location parameter δ_j.
- Since utility is T1EV and iid across alternatives the distribution of the max utility (the choice outcome) is given by

$$\begin{aligned}
\Pr\left(\max_{j\in\mathcal{J}} u_{ij} \leq u\right) &= \Pi_{j=1}^{J} \Pr(u_{ij} \leq u) \\
&= \Pi_{j=1}^{J} \exp\left(-\exp(-(u-\delta_j))\right) \\
&= \exp\left(-\textstyle\sum_{j=1}^{J} \exp(-(u-\delta_j))\right) \\
&= \exp\left(-\exp(-u)\textstyle\sum_{j=1}^{J} \exp(\delta_j)\right) \\
&= \exp\left(-\exp(-(u-w))\right) \\
&= G(u,w)
\end{aligned}$$

where $w = \log \sum_{j=1}^{J} \exp(\delta_j)$ (known as the log-sum term).

Simple logit: consumer surplus II

- From this we learn two useful things.
  1. The max-stability property of T1EV: The maximum of a set of iid T1EV draws is also T1EV (with location given by the log-sum term of their location parameters).
  2. Consumer surplus in logit has a closed form: Since the consumer surplus (CS) for a given consumer in the logit model is the maximum of a set of iid T1EV utility terms, by the max-stability property it has a T1EV distribution, with location given by the log-sum term of mean utilities. Hence, since the expectation of a T1EV is the location parameter plus the Euler constant we have

$$E\left[CS = \max_{j \in \mathcal{J}} u_{ij}\right] = \log \sum_{j=1}^{J} \exp(\delta_j) + 0.57722.$$

Note that this consumer surplus is in utils; to convert to currency units for welfare analysis we need to divide by the price parameter α.

The simple logit choice probability

- We now derive the logit choice probability. Consider $k \neq j$. Conditional on ε_{ij}: i prefers j to k with probability

$$\begin{aligned} \Pr\left(u_{ik} < u_{ij} \middle| \varepsilon_{ij} \right) &= \Pr\left(\varepsilon_{ik} < \delta_j - \delta_k + \varepsilon_{ij} \middle| \varepsilon_{ij} \right) \\ &= G(\delta_j - \delta_k + \varepsilon_{ij}) \end{aligned} \tag{27}$$

The simple logit choice probability

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- Conditional on ε_{ij}: the iid assumption implies i prefers j to k is independent of i prefers j to k' ($\forall k' \neq k$). So, to get the probability that i prefers j to all products in any set J' not containing j we can multiply together the individual probabilities for each product:

$$\Pr\left(u_{ik} < u_{ij}, \forall k \in J' \middle| \varepsilon_{ij}\right) = \prod_{k \in J'} G(\delta_j - \delta_k + \varepsilon_{ij}) \tag{28}$$

The simple logit choice probability

- Now stop conditioning on ε_{ij}: the unconditional choice probability follows by integration

$$
\begin{aligned}
&\Pr(u_{ik} < u_{ij}, \forall k \in J') \\
&= \int \left[\prod_{k \in J'} G(\delta_j - \delta_k + \varepsilon_{ij}) \right] g(\varepsilon_{ij}) \mathrm{d}\varepsilon_{ij} \\
&= \int \left[\prod_{k \in J'} \exp(-\exp(\delta_k - \delta_j - \varepsilon_{ij})) \right] \exp(-\varepsilon_{ij}) \exp(-\exp(-\varepsilon_{ij})) \mathrm{d}\varepsilon_{ij} \\
&= \frac{\exp(\delta_j)}{\exp(\delta_j) + \sum_{k \in J'} \exp(\delta_k)}
\end{aligned}
$$

substituting from (25) and (26) in 2nd line. See Train (2009) Ch3.10 for a derivation of final expression.

Simple logit: market share

- It follows that the market share of j in full choice set $\{0, \ldots, J\}$ is

$$s_j = \Pr(u_{ij} \geq u_{ik}, \forall k) = \frac{e^{\delta_j}}{\sum_{k \in \{0,\ldots,J\}} e^{\delta_k}} \tag{29}$$

  and that the market share of j in any subset $J_1 \subset \{0, \ldots, J\}$ containing j is

$$\Pr\left(u_{ij} \geq u_{ik} \ \forall k \in J_1\right) = \frac{e^{\delta_j}}{\sum_{k \in J_1} e^{\delta_k}}. \tag{30}$$

- The second result is useful later for the nested logit model.

Independence of Irrelevant Alternatives (IIA) property

- Ratio of market shares for any two products $(j, l) \in \{0, .., J\}$:

$$\frac{s_j}{s_l} = \frac{\exp \delta_j}{\sum_k \exp \delta_k} \Big/ \frac{\exp \delta_l}{\sum_k \exp \delta_k} = \frac{\exp \delta_j}{\exp \delta_l}.$$

  Note that this ratio is unaffected by

  1. Removal (or addition) of a product $k \notin \{j, l\}$. This enters via $(\sum_k \exp \delta_k)$ and cancels out.
  2. Increase (or reduction) of price for a product $k \notin \{j, l\}$. This enters via $(\sum_k \exp \delta_k)$ and cancels out.

- This is the IIA property. It implies that if product $k \notin \{j, l\}$ is removed, or its price increases, consumers substitute to products (j, l) in proportion to their market shares.

Estimation of simple logit model I

- To estimate we invert the market share function to obtain δ_j.
- To invert, log the ratio on previous slide:

$$\ln s_j - \ln s_l = (\delta_j - \ln \Sigma_{k=0}^{J} \exp \delta_k) - (\delta_l - \ln \Sigma_{k=0}^{J} \exp \delta_k)$$

which, since $\delta_0 = 0$, gives

$$\delta_j = \log s_j - \log s_0. \tag{31}$$

- Substituting $\delta_j = \beta x_j - \alpha p_j + \xi_j$ and rearranging gives

$$\xi_j = (\ln s_j - \ln s_0) - (\beta \boldsymbol{x}_j - \alpha p_j). \tag{32}$$

- Let $y_j = \log s_j - \log s_0$ and $\boldsymbol{x}_j^* = (\boldsymbol{x}_j, p_j)$. Then (32) is

$$y_j = \boldsymbol{x}_j^* \theta + \xi_j.$$

Estimation of simple logit model II

- Assume the $1 \times M$-vector $\boldsymbol{z}_j$ satisfies $E[\xi_j|\boldsymbol{z}_j] = 0$ which implies

$$E[\boldsymbol{z}_j'(y_j - \boldsymbol{x}_j^*\boldsymbol{\theta})] = \mathbf{0} \qquad (M \times 1).$$

- Let sample matrices be

$$\underset{(J\times M)}{\boldsymbol{Z}} = \begin{bmatrix} \boldsymbol{z}_1 \\ \vdots \\ \boldsymbol{z}_J \end{bmatrix}, \quad \underset{(J\times 1)}{\boldsymbol{y}} = \begin{bmatrix} y_1 \\ \vdots \\ y_J \end{bmatrix}, \quad \underset{(J\times dim(\boldsymbol{\theta}))}{\boldsymbol{X}} = \begin{bmatrix} \boldsymbol{x}_1 \\ \vdots \\ \boldsymbol{x}_J \end{bmatrix}.$$

- The corresponding sample moments are

$$(1/J)[\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})] \qquad (M \times 1) \tag{33}$$

- The GMM estimator is

$$\arg\min_{\boldsymbol{\theta}}[(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})'\boldsymbol{Z}]\boldsymbol{W}[\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})]$$

where W is an $M \times M$ weighting matrix.

Estimation of simple logit model III

- By solving the first order conditions

$$
\begin{aligned}
0 &= \frac{\partial}{\partial \boldsymbol{\theta}}[(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})'\boldsymbol{Z}]\boldsymbol{W}[\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})] \\
&= -2\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}[\boldsymbol{Z}'(\boldsymbol{y}\boldsymbol{X}\hat{\boldsymbol{\theta}})] \\
[\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{X}]\hat{\boldsymbol{\theta}} &= [\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{y}] \\
\hat{\boldsymbol{\theta}} &= [\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{X}]^{-1}[\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{y}]
\end{aligned}
$$

  the last line is the standard linear GMM IV estimator.
- For the standard errors as before $\widehat{Avar}(\hat{\boldsymbol{\theta}}) = \hat{\boldsymbol{A}}^{-1}\hat{\boldsymbol{B}}\hat{\boldsymbol{A}}^{-1}/J$, where

$$
\hat{\boldsymbol{A}} = \hat{\boldsymbol{G}}'\boldsymbol{W}\hat{\boldsymbol{G}}, \qquad \hat{\boldsymbol{B}} = \hat{\boldsymbol{G}}'\boldsymbol{W}\hat{\boldsymbol{V}}\boldsymbol{W}\hat{\boldsymbol{G}},
$$

  where for linear case

$$
\hat{\boldsymbol{G}} = -\boldsymbol{Z}'\boldsymbol{X}/J \qquad \hat{\boldsymbol{V}} = (1/J)\textstyle\sum_j(\boldsymbol{z}_j'(y_j - \boldsymbol{x}_j^*\hat{\boldsymbol{\theta}}))(\boldsymbol{z}_j'(y_j - \boldsymbol{x}_j^*\hat{\boldsymbol{\theta}})').
$$

- Extends to panel case where clustering on j may be desirable.

Weighting matrices and commonly used instruments

- Weighting matrix A popular (one-step) choice is 2SLS, which uses $\boldsymbol{W} = (\boldsymbol{Z}'\boldsymbol{Z})/J$. See Table 6.2, Cameron and Trivedi (2005).
- Instruments We later derive (42) that Nash equilibrium prices are

$$p_j = \text{m. cost}_j + \text{markup}_j \tag{34}$$

$$= c(\mathrm{w}_j) + \frac{1}{\alpha(1 - s_{f(j)})} \tag{35}$$

  where $s_{f(j)}$ is the share of the firm selling j.
- This suggests the following choice of instruments.
  - Marginal cost shifters Variables w_j excluded from utility so shift p_j only via c_j. $E(\xi_j|\mathrm{w}_j) = 0$.
  - Markup shifters e.g. number of products $\#J_{f(j)}$ sold by product j's firm, which should increase share $s_{f(j)}$. Excluded from utility, so $E(\xi_j|\#J_{f(j)}) = 0$.

Classwork 1: Generating data from a logit oligopoly model

Figure: Morrells choice beers, Malt Stout, Castle Ale and College Ale

Classwork I: Model primitives

- $T = 1000$, $J = 10$.
- Observed characteristics and unobserved utility:

$$\xi_{jt} \sim \mathrm{N}(0,1)$$
$$x_{jt} \sim \mathrm{Uniform}[0,1]$$

- Logit preferences with mean utility

$$\delta_{jt} = -1 + x_{jt} - p_{jt} + \xi_{jt}$$

- Marginal cost:

$$c_{jt} = x_{jt} + \omega_{jt} \qquad \omega_{jt} \sim \mathrm{Uniform}[0,1] \tag{36}$$

- Tri-opoly: each j assigned to $f = (1,2,3)$ w/ equal probability.
- Conduct: (i) marginal cost pricing or (ii) multi-product Nash.
- Note that j are iid across markets.

Classwork 1a: Generating market-level data and inversion

Use R code to carry out the following tasks:

1. Set seed. Fix the primitives and parameters of the model. Generate $(JT \times 1)$ vectors of product characteristics $\boldsymbol{x}_{jt}$, marginal costs $\boldsymbol{c}_{jt}$, unobserved utilities $\boldsymbol{xi}_{jt}$, market indices t, product indices j and firm ownership $f(jt)$ indices and collect in a data frame.
2. Write an R function `mkt_share()` for the market share function for market t: $\boldsymbol{s}_t = \boldsymbol{s}_t(\boldsymbol{p}_t, \boldsymbol{x}_t, \boldsymbol{xi}_t)$.
3. Generate a $(JT \times 1)$ vector of market shares assuming marginal cost pricing.
4. Invert the market share function and estimate the parameters of the model using OLS.

Code is in classwork1a.R

Task 1: fix primitives and parameters

```
J <- 10                          #products
T <- 1000                        #markets
N <- T*J                         #observations
f <- sample(c(1, 2, 3), size = N, replace = TRUE)
t <- rep(1:T, each=J)
tf <- as.numeric(interaction(f, t, drop = TRUE))

xj  <- runif(N)                  #x
xi <- rnorm(N)              #xi
mc <- xj + runif(N)         #mc

exog.data <- data.frame(t, f, tf, xj, mc, xi)

alpha <- 1                      #parameter on p
beta1 <- -1                     #constant
beta2 <- 1                      #parameter on x

theta <- c(alpha,beta1,beta2)
```

Task 2: market share function (t-specific)

```
#Arguments: prices/data for market t and parameters.

mkt_share <- function(price, exog.data_t,  theta) {

    x = exog.data_t$xj        #Jx1 characteristics
    xi = exog.data_t$xi       #Jx1 unobserved utility

    u = theta[2] + theta[3]*x - theta[1]*price + xi

    numerator <- exp(u)
    denom  <- 1+sum(numerator)

    return (numerator / denom) #Jx1 market shares
}
```

Task 3: Get market shares for all t

```
sj_all = c() #for concat. (JT x 1) sj (inside)
s0_all = c() #for concat. (JT x 1) s0 (outside)
pj_all <- mc #marginal cost pricing

for (k in 1:T) {
    pj_t <- pj_all[ t == k ] #select k
    exog.data_t = exog.data[t == k ,]

    sj_t <- mkt_share(pj_t, exog.data_t, theta)
    s0_t <- rep(1-sum(sj_t),J)

    pj_all = c(pj_all, pj_t)
    sj_all = c(sj_all, sj_t)
    s0_all = c(s0_all, s0_t)
}
```

Task 4: Inversion and OLS

```
# Invert market share
delta = log(sj_all)-log(s0_all)


#OLS
model_ols <- lm(delta~xj+pj_all)
summary(model_ols)

#Check covariance
cor(pj_all,xi)
```

Logit Elasticities and Endogenous Prices I

- Two useful derivatives. From (29) we get the own-price effect:

$$\begin{aligned}\frac{\partial s_j}{\partial p_j} &= \frac{\partial}{\partial \delta_j}\left(\frac{\exp \delta_j}{\sum_k \exp \delta_k}\right)\frac{\partial \delta_j}{\partial p_j} \\ &= \left(\frac{\exp \delta_j}{\sum_k \exp \delta_k} - \frac{\exp \delta_j \exp \delta_j}{\left(\sum_k \exp \delta_k\right)^2}\right)(-\alpha) \\ &= -\alpha(1 - s_j)s_j \end{aligned} \tag{37}$$

and the cross-price effect:

$$\begin{aligned}\frac{\partial s_l}{\partial p_j} &= \frac{\partial}{\partial \delta_j}\left(\frac{\exp \delta_l}{\sum_k \exp \delta_k}\right)\frac{\partial \delta_j}{\partial p_j} \\ &= -\frac{\exp \delta_j \exp \delta_l}{\left(\sum_k \exp \delta_k\right)^2}(-\alpha) \\ &= \alpha s_j s_l. \end{aligned} \tag{38}$$

Simple logit elasticities I

- Thus own-price elasticities are

$$\frac{p_j}{s_j}\frac{\partial s_j}{\partial p_j} = -\alpha p_j(1 - s_j), \tag{39}$$

cross-price elasticities are

$$\frac{p_j}{s_l}\frac{\partial s_l}{\partial p_j} = \alpha p_j s_j, \tag{40}$$

and nothing in the RHS of (56) is l-specific so elasticity (with respect to p_j) same for all products bar j.

- Putting it another way: When product j changes price there are only two values for elasticities: one for j and one for all other products.

Simple logit markups and marginal costs I

- Single-product profit is $\Pi_j = (p_j - c_j)s_j$
- Assuming single-product Nash implies $\partial \Pi_j / \partial p_j = 0$, $\forall j$.
- Using (37) this condition implies markups are

$$\begin{aligned} p_j - c_j &= s_j \left[\partial s_j / \partial p_j\right]^{-1} \\ &= \frac{1}{\alpha(1 - s_j)} \end{aligned}$$

  and marginal costs are

$$c_j = p_j - \frac{1}{\alpha(1 - s_j)}.$$

  Hence marginal cost is identified if α is known.
- Lerner index (market power) is inverse of own-price elasticity

$$\frac{p_j - c_j}{p_j} = -\left[\frac{p_j}{s_j}\frac{\partial s_j}{\partial p_j}\right]^{-1} = \frac{1}{p_j}\frac{1}{\alpha(1 - s_j)}$$

Simple logit markups and marginal costs II

- Now suppose multi-product firm f owns product set J_f.
- Profit of multi-product firm f is $\Pi_f = \sum_{j \in J_f}(p_j - c_j)s_j$.
- Multi-product Nash implies $\partial \Pi_f / \partial p_j = 0$, $\forall j \in J_f$ and

$$\begin{aligned} 0 &= s_j + \sum\nolimits_{l \in J_f}(p_l - c_l)\frac{\partial s_l}{\partial p_j} \\ &= s_j + \alpha \sum\nolimits_{l \in J_f}(p_l - c_l)s_j s_l - \alpha(p_j - c_j)s_j \end{aligned}$$

using (37,38). Rearranging we get markups

$$(p_j - c_j) = \frac{1}{\alpha} + \sum_{l \in J_f}(p_l - c_l)s_l \tag{41}$$

which implies $(p_j - c_j)$ is same for all $j \in J_f$ so that

$$(p_j - c_j) = \frac{1}{\alpha(1 - s_f)} \tag{42}$$

where $s_f = \sum_{l \in J_f} s_l$. Hence, markups (and market power) depend on firm's market share and c_j is identified.

Simple logit markups and marginal costs III

- For the two pricing models considered – single-product Nash and multi-product Nash – we can write the problem as follows

$$\boldsymbol{p}_t = \boldsymbol{c}_t + \boldsymbol{\eta}_t(\boldsymbol{s}_t(\boldsymbol{p}_t)).$$

  where $\boldsymbol{\eta}_t$ is the markup term (different for each model).
- This is a nonlinear system of J equations in J unknown prices which can be solved numerically either by derivative based (Newton) methods or (if it is a contraction) by iteration.
- For example in R we can use `nleqslv()` (which uses the Newton method) to find prices $\boldsymbol{p}_t$ such that

$$\boldsymbol{p}_t - (\boldsymbol{c}_t + \boldsymbol{\eta}_t(\boldsymbol{s}_t(\boldsymbol{p}_t))) = \mathbf{0}.$$

Classwork 1b: Endogenous prices, instruments, elasticities

Continuing from Classwork 1a, use R code to carry out the following tasks:

1. Write an R function `foc_logit()` where multi-product Nash pricing is assumed that takes a trial set of prices $\boldsymbol{p}_t$ and returns the value
$$\boldsymbol{p}_t - (\boldsymbol{c}_t + \boldsymbol{\eta}_t(\boldsymbol{s}_t(\boldsymbol{p}_t))).$$
which is zero at equilibrium prices.
2. Use `nleqslv` to solve for equilibrium prices in each market. Concatenate the prices and market shares in $(JT \times 1)$ vectors.
3. Invert the market share function and estimate the parameters of the model using OLS and 2SLS. Run two 2SLS verions with the following instruments for price respectively: marginal costs and sum of same-firm same-market product characteristics.
4. Calculate $J \times J$ matrix of own- and cross-price elasticities for market 1 and assess flexibility.

Code is in classwork1b.R

Task 1: system of Nash prices (t-specific)

```
foc_logit <- function(price, exog.data_t) {

  mc <- exog.data_t$mc
  f <- exog.data_t$f

  s <- mkt_share(price,exog.data_t,  theta) #(Jx1)

  #Gen t-specific index (some t don't have all 3 firms)
  y <- match(f,unique(f))

  #Aggregate shares to f-level
  agg <- as.numeric(tapply(s, y, sum))   # (Fx1)
  sf <- agg[y]                           # (Jx1)

  markup <- 1/(1-sf)
  return(price - mc - markup)  #0 iff satisfy FOCs (Jx1)
}
```

Task 2: Solve for p_t and s_t in each t and concatenate

```
pj_all = c() #for concat. (JT x 1) prices
sj_all = c() #for concat. (JT x 1) sj (inside)
s0_all = c() #for concat. (JT x 1) s0 (outside)

for (k in 1:T) {
    p0_t <- 1.25*mc[ t == k ] #starting guess
    exog.data_t = exog.data[t == k ,]

    Y<-nleqslv(p0_t,foc_logit,jac=NULL, exog.data_t,theta)

    p1_t <- Y$x #solved unknown stored as x
    sj_t <- mkt_share(p1_t, exog.data_t, theta)
    s0_t <- rep(1-sum(sj_t),J)

    pj_all = c(pj_all, p1_t)
    sj_all = c(sj_all, sj_t)
    s0_all = c(s0_all, s0_t)
}
```

Task 3: Invert and Estimate

```
delta = log(sj_all)-log(s0_all)

model_ols <- lm(delta~xj+pj_all) #OLS
summary(model_ols)
cor(pj_all,xi)

model_iv <- ivreg(delta~xj|pj_all|mc) #IV: cost-shifters
summary(model_iv)

# Sum/count xj for rows with the same tf:
temp <- as.numeric(tapply(xj, tf, sum))
BLP_sumx <- temp[tf]

model_iv <- ivreg(delta~xj|pj_all|BLP_sumx) #IV: BLP
summary(model_iv)
```

Task 4: Elasticity Matrix

```
a_hat = -model_iv$coefficients[2] #sgn to +ve as notation
mkt_ind <- t == 1
sj_t = sj_all[mkt_ind]
pj_t = pj_all[mkt_ind]

# Get JxJ cross-elasticities
sp_mkt <- a_hat*pj_t*sj_t # (J x 1) j's price changes
CrossElas = matrix(sp_mkt, nrow=J,ncol=J)

# Transpose matrix so get row's s wrt column's p
CrossElas = t(CrossElas)

#Get own-price elasticity vector and put it on diagonal:
ownElas <- -a_hat*pj_t*(1-sj_t)
diag(CrossElas) <- ownElas
round(CrossElas,3)
```

Classwork 1: discussion I

- Endogenous demand and prices
- Direction of bias
- IV; alternative choices
- Elasticities: flexibility and IIA
- Elasticities: relationship to price
- Standard errors.

Figure: Ford Sierra Sapphire

Application: BLP (1995) Autos Model, simple logit case I

- BLP (1995) begin their paper with a simple logit model.
- The data dimensions are as follows: 26 firms (f), 999 models (j), 20 years ($t = 1970 - 1990$), and 2217 product-year observations (j, t).
- p_{jt} is the list price (in \$1983).
- s_{jt} is the quantity of j sold in year t divided by total number of US households (from US census) in year t. Hence, $s_{0t} = 1 - \Sigma_{j>0} s_{jt}$.
- There are 5 characteristics $\boldsymbol{x}_j = (x_{1j}, , x_{5j})$:
  - $x_{1j} = 1$,
  - x_{2j} is j's HP divided by weight (hpwt),
  - x_{3j} is air conditioning dummy (air),
  - x_{4j} is length times width (size),
  - x_{5j} is MP\$ (miles per dollar, mpd).

Application: BLP (1995) Autos Model, simple logit case II

- Simple logit form model in linear form by rearranging (32):

$$(\log s_{jt} - \log s_{0t}) = \beta \boldsymbol{x}_j - \alpha p_{jt} + \xi_{jt} \tag{43}$$

- BLP assume that ξ_{jt} is revealed to firms after they set characteristics $\boldsymbol{x}_j$ but before they set prices p_{jt} and hence that we can use $\boldsymbol{x}_j$ but not p_{jt} in $\boldsymbol{z}_j$.
- The 15 (demand side) instruments BLP use are:

$$\boldsymbol{z}_j = (\boldsymbol{x}_j, \boldsymbol{x}_j^{\text{samef}}, \boldsymbol{x}_j^{\text{otherf}}) \tag{44}$$

where

$$\begin{aligned}
\boldsymbol{x}_j^{\text{samef}} &= (\textstyle\sum_{\{l:l\neq j, l\in\mathcal{J}_{f(j)}\}} x_{1l}, \ldots, \sum_{\{l:l\neq j, l\in\mathcal{J}_{f(j)}\}} x_{5l}) \\
\boldsymbol{x}_j^{\text{otherf}} &= (\textstyle\sum_{\{l:l\notin\mathcal{J}_{f(j)}\}} x_{1l}, \ldots, \sum_{\{l:l\notin\mathcal{J}_{f(j)}\}} x_{5l})
\end{aligned}$$

where $\mathcal{J}_{f(j)}$ is products produced by firm f that produces j.

Application: BLP (1995) Autos Model, simple logit case III

- $\boldsymbol{x}_j$ are informative about β but why are the other instruments informative about α?
- The instruments marked "samef" and "otherf" are sometimes called market structure instruments. In particular since $x_{1j} = 1$, the first element in x_j^{samef} and x_j^{otherf} are product ownership counts. Why might these instruments be related to p_j?
- The answer is that they enter the supply side: Nash equilibrium prices for the logit model, in (42), are increasing in the market share of the firm s_f and decreasing in the market share of the other firms. Hence,
  - products tend to have low markups if they face better/more substitutes from rival firms and x_j^{otherf} measures this;
  - products tend to have high markups if products that face better/more substitutes supplied by the product's own firm and x_j^{samef} measures this.
- There are $M = 15$ instruments to estimate 6 parameters.
- BLP estimate by 2SLS, clustering by product.

Application: BLP (1995) Autos Model, simple logit case IV

- The classwork2 session (see classwork2.pdf on course Canvas site) estimates a simple logit model using the inverted market share function.
- Linear regression by OLS and 2SLS.
- Data: Auto demand from BLP(1995).
- IVs: BLP instruments.
- Post-estimation: elasticities, marginal costs, markups.

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More flexible alternatives to logit

- We now consider three alternatives to logit that are more flexibile.
- These fall into two groups:
  - Models where ε_{ijt} is not iid across products: probit, nested logit.
  - Models where tastes for product characteristics β vary across consumers: mixed logit.
- We begin with nested logit which has a closed form for market share and its inverse.
- The other two (probit, mixed logit) require market shares to be simulated. We will discuss probit very briefly and focus on mixed logit.

Nested logit

- Nesting of Products Suppose that there are product groups

$$r = 1, \ldots, R$$

  within which products are closer substitutes
- Let $\mathcal{J}_r$ be the set of products in group r.
- Each product j is in only one group.
- ε_{ij} positively correlated for products in same group; iid for products in different groups, and iid for two products in same group after conditioning on being in the group.
- Allows products within a group to be 'closer substitutes' than products in different groups – i.e. people who have a high taste draw for one product in a group are more likely to have a high taste draw for other products in the same group.

Generalized Extreme Value (GEV) distribution

- GEV allows positive correlation between ε_{ij} for $j \in \mathcal{J}_r$.
- The GEV distribution function:

$$G(\varepsilon_{i1}, ..., \varepsilon_{iJ}) = \exp\left(-\sum_r \left(\sum_{j \in \mathcal{J}_r} \exp(-\varepsilon_{ij}/(1-\sigma))\right)^{1-\sigma}\right). \tag{45}$$

  where $\sigma \in (0, 1)$.
- σ can vary by r but we suppress that to keep notation simple.
- Reduces to simple logit when $\sigma = 0$: this is because $G(\varepsilon_{i1}, ..., \varepsilon_{iJ})$ becomes a product of the Type 1 EV distribution function: $G(\varepsilon_{i1}, ..., \varepsilon_{iJ}) = G(\varepsilon_{i1}) \times ... \times G(\varepsilon_{iJ})$ where $G(\varepsilon) = \exp(-\exp(-\varepsilon))$.

An intuitive decomposition (Berry (1994))

- A GEV disturbance ε_{ij} can decomposed into additive error-component form as follows:

$$\varepsilon_{ij} = \varepsilon_{ir}^{(1)} + (1-\sigma)\varepsilon_{ij}^{(2)}$$

  - $\varepsilon_{ir}^{(1)}$ is i's taste disturbance for group r.
  - $\varepsilon_{ij}^{(2)}$ is Type-1 EV (as in simple logit).
- Substituting this into utility we get

$$u_{ij} = \delta_j + \varepsilon_{ir}^{(1)} + (1-\sigma)\varepsilon_{ij}^{(2)} \tag{46}$$

- Note: $\varepsilon_{ir}^{(1)}$ does not affect preference orderings *within* group r.
- Can divide all utilities (46) by $(1-\sigma)$ without affecting rankings.
- Hence, as $\varepsilon_{ij}^{(2)}$ is Type-1 EV the probability of choice within a group is simple logit (see argument for equation (30))

$$(s_j/s_r) = \Pr\left(u_{ij} > u_{ik}\ \forall k \in \mathcal{J}_r\right) = \frac{\exp\left(\delta_j/(1-\sigma)\right)}{\sum_{k\in\mathcal{J}_r}\exp\left(\delta_k/(1-\sigma)\right)}. \tag{47}$$

Market share of group and product

- It can be shown that group share s_r is

$$s_r = \frac{D_r^{(1-\sigma)}}{\sum_{r'} D_{r'}^{(1-\sigma)}} \text{ where } D_r = \sum_{k \in \mathcal{J}_r} \exp\left(\delta_k/(1-\sigma)\right).$$

- Thus overall share of j is a product of two logits

$$s_j = (s_j/s_r)s_r$$

$$= \underbrace{\frac{\exp\left(\delta_j/(1-\sigma)\right)}{\sum_{k \in \mathcal{J}_r} \exp\left(\delta_k/(1-\sigma)\right)}}_{\text{within-group r choice prob.}} \underbrace{\frac{D_r^{(1-\sigma)}}{\sum_{r'} D_{r'}^{(1-\sigma)}}}_{\text{prob. choose group r}} . \tag{48}$$

- The outside good, with $\delta_0 = 0$, is alone in group $r = 0$, so $D_0 = 1$.

A relaxation of IIA

1. For two products j and l in the same group r, IIA holds:

$$\frac{s_j}{s_l} = \frac{\frac{\exp(\delta_j/(1-\sigma))}{D_r}}{\frac{\exp(\delta_l/(1-\sigma))}{D_r}} \times \frac{s_r}{s_r} = \frac{e^{\delta_j/(1-\sigma)}}{e^{\delta_l/(1-\sigma)}}$$

2. For products in different groups (r and $r \neq r'$), IIA does not hold:

$$\frac{s_j}{s_l} = \frac{\frac{\exp(\delta_j/(1-\sigma))}{D_r}}{\frac{\exp(\delta_l/(1-\sigma))}{D_{r'}}} \times \frac{s_r}{s_{r'}} \tag{49}$$

Case of $l = 0$ (for inversion): setting $\delta_0 = 0$ and $D_0 = 1$ in (49) gives

$$\frac{s_j}{s_0} = \frac{\exp\left(\delta_j/(1-\sigma)\right)}{D_r}\frac{s_r}{s_0} = \frac{\exp\left(\delta_j/(1-\sigma)\right)}{D_r} D_r^{(1-\sigma)} \tag{50}$$

where second equation is by cancelling common denominators in group share expressions.

Estimation of nested logit model I

- To invert we log and apply algebra:

$$\begin{aligned}\ln s_j - \ln s_0 &= \delta_j/(1-\sigma) - \sigma \ln D_r \\ &= \delta_j + \sigma\delta_j/(1-\sigma) - \sigma \ln D_r \\ &= \delta_j + \sigma[\delta_j/(1-\sigma) - \ln D_r] \\ &= \delta_j + \sigma \ln(e^{(\delta_j/(1-\sigma))}/D_r) \\ &= \delta_j + \sigma \ln(s_j/s_r)\end{aligned}$$

- Substituting in $\delta_j = \beta \boldsymbol{x}_j - \alpha p_j + \xi_j$:

$$\xi_j = (\log s_j - \log s_0) - \sigma \ln(s_j/s_r) - (\beta \boldsymbol{x}_j - \alpha p_j). \tag{51}$$

- Let $y_j = \log s_j - \log s_0$ and $\boldsymbol{x}_j^* = (\ln(s_j/s_r), \boldsymbol{x}_j, p_j)$ so that (51) is

$$y_j = \boldsymbol{x}_j^* \theta + \xi_j.$$

Estimation of nested logit model II

- Assume the $1 \times M$-vector $\boldsymbol{z}_j$ satisfies $E[\xi_j|\boldsymbol{z}_j] = 0$ which implies

$$E[\boldsymbol{z}_j'(y_j - \boldsymbol{x}_j^*\boldsymbol{\theta})] = \mathbf{0} \qquad (M \times 1).$$

- Let sample matrices be

$$\begin{array}{ll} \boldsymbol{Z} & (J \times M) \\ \boldsymbol{y} & (J \times 1) \\ \boldsymbol{X} & (J \times dim(\boldsymbol{\theta})) \end{array}$$

so corresponding sample moments are

$$(1/J)[\boldsymbol{Z}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})] \qquad (M \times 1) \tag{52}$$

and as with simple logit we use the linear GMM IV estimator

$$\hat{\theta} = [\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{X}]^{-1}[\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{Y}]. \tag{53}$$

where W is an $M \times M$ weighting matrix.

Estimation of nested logit model III

- Instruments. Consider the error

$$\xi_j = (\log s_j - \log s_0) - \sigma \ln(s_j/s_r) - (\beta \boldsymbol{x}_j - \alpha p_j). \tag{54}$$

- Compared to logit, an extra endogenous variable: $\ln((s_j/s_r))$.
- A possible choice for the vector of instruments is

$$\boldsymbol{z_j} = \{\boldsymbol{x}_j, c_j, \#J_{r(j)}\}_j$$

where

- $\boldsymbol{x}_j$ (product characteristics) instruments for themselves.
- c_j (marginal cost) correlated with p_j
- $\#J_{r(j)}$ (number of products in j's group) measures whether j is crowded or isolated in its group and is correlated with $\ln(s_j/s_r)$.

Price elasticities, marginal costs, flexibility I

- The own-price elasticity for product j in group r is

$$\frac{p_j}{s_j}\frac{\partial s_j}{\partial p_j} = -\frac{\alpha}{1-\sigma} p_j [1 - \sigma(s_j/s_r) - (1-\sigma)s_j] \tag{55}$$

and cross-price elasticities are

$$\frac{p_j}{s_l}\frac{\partial s_l}{\partial p_j} = \begin{cases} \alpha p_j s_j \left(1 + \frac{\sigma}{1-\sigma} s_r\right) & (l,j) \text{ are in nest } r \\ \alpha p_j s_j & (l,j) \text{ are in different nests} \end{cases}. \tag{56}$$

- Assuming single-product Nash, markups (by inverse-elasticity pricing rule) are available from (55):

$$p_j - c_j = s_j \left[\partial s_j / \partial p_j\right]^{-1} \tag{57}$$

and marginal costs are identified. If there is multi-product pricing then a more complicated expression should be used.

Price elasticities, marginal costs, flexibility II

- Click on Miller-Weinberg notes for derivations of elasticities.
- How much more flexible is this? For a change in p_j we now have two distinct cross-elasticity values (given on previous slide) and logit has one. So it is more flexible than logit but not much more. This is not surprising: we have only added one parameter (or one parameter per nest if we allow σ to vary by nest).
- To add more flexibility the nested logit can have more levels of nesting (three-level, four-level, etc., nested logit), e.g. for car choice: (i, top nest) inside goods or outside good, (ii, middle nest) then choose foreign or domestic, (iii, bottom nest) then choose SUV or sports cars, then choose model. This is done in Goldberg (Econometrica, 1995) using consumer-level data. See Train's (2009) book for discussion of the method.

Classwork 3: Generating data from a nested logit model

Figure: Morrells choice beers, Malt Stout, Castle Ale and College Ale

Classwork 3 I

- $T = 1000$ markets, $J = 10$ products, $R = 2$ nests + outside good
- Observed characteristics and unobserved utility:

$$\xi_{jt} \sim 0.5 * \mathrm{N}(0,1)$$
$$x_{jt} \sim 0.5 * \mathrm{Uniform}[0,1]$$

- Nested-logit (GEV) preferences with mean utility

$$\delta_{jt} = -2 + x_{jt} - p_{jt} + \xi_{jt} \qquad \delta_{0t} = 0$$

- Nesting parameter $\sigma = 0.5$
- Marginal cost:

$$c_{jt} = x_{jt} + 0.5 * \omega_{jt} \qquad \omega_{jt} \sim 0.5 * \mathrm{Uniform}[0,1] \tag{58}$$

- j allocated to group $r = (1,2)$ randomly (allocation iid across markets).

Classwork 3 II

- Conduct: (single-product-firm) Nash pricing.
- Generate endogenous prices and market shares with Nested logit market shares and single-product Nash pricing using a nonlinear equation solver.
- Invert market share system analytically and estimate by OLS.
- Estimate by IV instrumenting only for price (using marginal cost); show this is not sufficient.
- Generate the instrument for the within-group market share: a count of the number of goods in each market-group.
- Estimate by IV.
- Calculate an own- and cross-price elasticity matrix to assess additional flexibility relative to simple logit.

See Classwork3.pdf and suggested code on Canvas.

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Probit

- Model as simple logit except $(\varepsilon_{i1}, ..., \varepsilon_{iJ}) \sim N(\mathbf{0}, \boldsymbol{\sigma})$ (multivariate normal), where $\boldsymbol{\sigma}$ is a $J \times J$ variance co-variance matrix (elements σ_{jl}), which allows free correlation.
- The probability of choosing j is

$$s_j(\boldsymbol{\delta}, \boldsymbol{\sigma}) = \int 1(\delta_j + \varepsilon_{ij} > \delta_l + \varepsilon_{il},\ \forall l \neq j) g(\boldsymbol{\varepsilon}, \boldsymbol{\sigma}) \mathrm{d}\boldsymbol{\varepsilon}$$

  where g is multivariate Normal density.
- Advantage: flexible, as the covariances are free.
- Disadvantages:
  - number of parameters in $\boldsymbol{\sigma}$ is J^2 which may be too many when J is large;
  - no analytical form for market share, have to compute by simulation, $s_j^{sim}(\boldsymbol{\delta}, \boldsymbol{\sigma})$, difficult when J is more than a few; and
  - no analytical form for $\delta(\boldsymbol{s}, \boldsymbol{\sigma})$ and no contraction proof.
- Used only when J is a few, e.g. Goolsbee-Petrin (2004).
- From now on we do mixed logit.

Mixed logit utility

- With mixed logit ε_{ij} are iid Type-1 EV and we have heterogeneity in tastes for characteristics/price so that

$$u_{ij} = \sum_k (\beta_k + \sigma_k \nu_{ik}) x_{jk} - (\alpha + \sigma_0 \nu_{i0}) p_j + \xi_j + \varepsilon_{ij}$$

  where ν_{ik} and ν_{i0} are iid across consumers from distribution H.
- H can be a $N(0,1)$ or the empirical distribution of some demographic attribute.
- σ_k and σ_0 are scaling terms that allow heterogeneity of tastes in the population to vary by characteristic k.
- Collecting the mean utility terms

$$u_{ij} = \delta_j + \sum_k \sigma_k \nu_{ik} x_{jk} - \sigma_0 \nu_{i0} p_j + \varepsilon_{ij} \tag{59}$$

$$\delta_j = \sum_k \beta_k x_{jk} - \alpha p_j + \xi_j \tag{60}$$

  Set $\delta_0 = 0$ for $j = 0$ so that $u_{i0} = \varepsilon_{i0}$.

Mixed logit market share function

Conditional on $\boldsymbol{\nu_i}$, this is simple logit:

$$s_{ij} = s_j(\boldsymbol{\delta}, \boldsymbol{\sigma}, \boldsymbol{\nu_i}) = \frac{\exp\left(\delta_j + \sum_k \sigma_k \nu_{ik} x_{jk} - \sigma_0 \nu_{i0} p_j\right)}{1 + \sum_{l=1}^{J} \exp\left(\delta_l + \sum_k \sigma_k \nu_{ik} x_{lk} - \sigma_0 \nu_{i0} p_l\right)}$$

Integrate out ν to get market share function

$$s_j(\boldsymbol{\delta}, \boldsymbol{\sigma}) = \int \frac{\exp\left(\delta_j + \sum_k \sigma_k \nu_k x_{jk} - \sigma_0 \nu_0 p_j\right)}{1 + \sum_{l=1}^{J} \exp\left(\delta_l + \sum_k \sigma_k \nu_k x_{lk} - \sigma_0 \nu_0 p_l\right)} \mathrm{d}H(\nu)$$

No analytical form so simulate, with ns draws from H

$$s_j^{sim}(\boldsymbol{\delta}, \boldsymbol{\sigma}) = \frac{1}{ns} \sum_{i=1}^{ns} \frac{\exp\left(\delta_j + \sum_k \sigma_k \nu_{ik} x_{jk} - \sigma_0 \nu_{i0} p_j\right)}{1 + \sum_{l=1}^{J} \exp\left(\delta_l + \sum_k \sigma_k \nu_{ik} x_{lk} - \sigma_0 \nu_{i0} p_l\right)}$$

Theoretical Properties of Mixed Logit

- How much more flexible than logit?
- Depends on σ.
- If $\sigma = 0$ for all σ terms then we return to simple logit.
- If $\sigma > 0$ for price, then we expect that people who like one low-price car also tend to like another low-price car.
- This breaks the IIA property: if a low-price car increases its price (or is withdrawn) then consumers will switch disproportionately to other low-price cars. And, if the other σ terms are positive, to other cars with similar characteristics.
- The demand derivatives are now simulated:

$$\frac{\partial s_j}{\partial p_l} = \frac{1}{ns} \sum_{i=1}^{ns} \frac{\partial s_{ij}}{\partial p_l}$$

where the derivative conditional on type is as for simple logit.

Inversion of market share: mixed logit

- In mixed logit there's no analytical form for $\boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{\sigma})$.
- Can solve for $\boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{\sigma})$ numerically. BLP (1995) show that the following is a contraction mapping

$$\boldsymbol{\delta}^{(n)} = \boldsymbol{\delta}^{(n-1)} + \ln \boldsymbol{s} - \ln \boldsymbol{s}^{sim}(\boldsymbol{\delta}^{(n-1)}, \boldsymbol{\sigma}) \tag{61}$$

  where $\boldsymbol{s}$ is the observed market shares.
- A contraction mapping $\boldsymbol{g}(\boldsymbol{\delta})$ has a unique fixed point (i.e. one point where $\boldsymbol{\delta} = \boldsymbol{g}(\boldsymbol{\delta})$) to which (by the Banach fixed-point theorem) the iterated function sequence $\{\boldsymbol{\delta}, \boldsymbol{g}(\boldsymbol{\delta}), \boldsymbol{g}(\boldsymbol{g}(\boldsymbol{\delta})), \boldsymbol{g}(\boldsymbol{g}(\boldsymbol{g}(\boldsymbol{\delta}))), \ldots\}$ converges.
- Thus $\boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{\sigma})$ may be solved by iteration starting at an initial guess for $\boldsymbol{\delta}^{(0)}$ in (61) and updating $\boldsymbol{\delta}^{(n)}$ until the difference between $\boldsymbol{\delta}^{(n)}$ and $\boldsymbol{\delta}^{(n-1)}$ is very small, e.g. the stopping rule

$$\left\| \boldsymbol{\delta}^{(n)} - \boldsymbol{\delta}^{(n-1)} \right\| \leq 1 \times e^{-09}.$$

- The solution to this iteration is $\boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{\sigma})$.

GMM estimation of mixed logit I

- We rewrite ξ_j for $\boldsymbol{\theta} = (\boldsymbol{\sigma}, \boldsymbol{\beta}, \alpha)$

$$\xi_j(\boldsymbol{\theta}) = \delta_j(\boldsymbol{s}, \boldsymbol{\sigma}) - (\boldsymbol{\beta}\boldsymbol{x}_j - \alpha p_j).$$

- Assume a vector of M instruments $\boldsymbol{z}$ satisfies the population moment condition $E[\xi|\boldsymbol{z}] = 0$ which, with $E[\xi] = 0$, implies

$$E[z_m \xi] = 0 \text{ for } m = 1, \ldots, M.$$

- The corresponding sample moments are

$$g_m(\boldsymbol{\theta}) = \tfrac{1}{J}\textstyle\sum_{j=1}^{J} \xi_j(\boldsymbol{\theta}) z_{m,j}. \tag{62}$$

- Stack the M moments to get the $M \times 1$ vector

$$\boldsymbol{g}(\boldsymbol{\theta}) = \begin{bmatrix} g_1(\boldsymbol{\theta}) \\ \vdots \\ g_M(\boldsymbol{\theta}) \end{bmatrix}$$

GMM estimation of mixed logit II

- Given the $M \times M$ weighting matrix $\boldsymbol{W}$, GMM estimates are

$$\arg\min_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}) = \boldsymbol{g}(\boldsymbol{\theta})'\boldsymbol{W}\boldsymbol{g}(\boldsymbol{\theta}) \tag{63}$$

- As the covariance moments $\boldsymbol{g}(\boldsymbol{\theta})$ use errors

$$\xi_j(\boldsymbol{\theta}) = \delta_j(\boldsymbol{s}, \boldsymbol{\sigma}) - (\beta \boldsymbol{x}_j - \alpha p_j),$$

  that are nonlinear in (some of the) parameters, must minimize this numerically rather than use linear GMM IV estimator.
- Since errors are linear in (β, α) the linear GMM IV estimators

$$(\hat{\beta}(\boldsymbol{\sigma}), \hat{\alpha}(\boldsymbol{\sigma})) = [\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\boldsymbol{X}]^{-1}[\boldsymbol{X}'\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}'\delta(\boldsymbol{\sigma})]. \tag{64}$$

  minimize Q for any given $\boldsymbol{\sigma}$, where $\boldsymbol{X}$ is the matrix of $(\boldsymbol{x}_j, p_j)$ observations and $\boldsymbol{Z}$ is the $J \times M$ matrix of instruments.
- This is called concentrating out the linear parameters: we need only search *numerically* for $\boldsymbol{\sigma}$, reducing computation costs.

GMM estimation of mixed logit III

- Concentrating out. Given any $\boldsymbol{\sigma}$ (and implied linear parameters):

$$\xi_{jt}(\boldsymbol{\sigma}) = \delta_j(\boldsymbol{s}, \boldsymbol{\sigma}) - \hat{\beta}(\boldsymbol{\sigma})\boldsymbol{x}_j + \hat{\alpha}(\boldsymbol{\sigma})p_j,$$

so that the sample moments are (for each m)

$$g_m(\boldsymbol{\sigma}) = \tfrac{1}{J}\textstyle\sum_{j=1}^{J}\xi_j(\boldsymbol{\sigma})z_{m,j}. \tag{65}$$

the stacked $M \times 1$ vector is

$$\boldsymbol{g}(\boldsymbol{\sigma}) = \begin{bmatrix} g_1(\boldsymbol{\sigma}) \\ \vdots \\ g_M(\boldsymbol{\sigma}) \end{bmatrix}$$

and GMM estimator is

$$\arg\min_{\boldsymbol{\sigma}} Q(\boldsymbol{\sigma}) = \boldsymbol{g}(\boldsymbol{\sigma})'\boldsymbol{W}\boldsymbol{g}(\boldsymbol{\sigma}) \tag{66}$$

which is a lower-dimensional search problem than (63).

Summarising the Nested Fixed Point (NFXP) algorithm

1. Start with some initial value of $\boldsymbol{\theta} = (\boldsymbol{\sigma}, \beta, \alpha)$
2. At trial value $\boldsymbol{\sigma}$ obtain $\boldsymbol{\delta}(\boldsymbol{\sigma})$ by iterating

$$\boldsymbol{\delta}_n = \boldsymbol{\delta}_{n-1} + \ln \boldsymbol{s} - \ln \boldsymbol{s}^{sim}(\boldsymbol{\delta}_{n-1}, \boldsymbol{\sigma})$$

   to convergence. In this, $\boldsymbol{s}$ is observed, and $\boldsymbol{s}^{sim}(\boldsymbol{\delta}_{n-1}, \boldsymbol{\sigma})$ is simulated (has to be done at each iteration).
3. Concentrate out the linear parameters $(\hat{\beta}, \hat{\alpha})$ implied by $\boldsymbol{\sigma}$.
4. Compute $\xi_j(\boldsymbol{\sigma}) = \delta_j(\boldsymbol{s}_t, \boldsymbol{\sigma}) - \hat{\beta}\boldsymbol{x}_j - \hat{\alpha}p_j$ for all (j).
5. Evaluate $\boldsymbol{g}(\boldsymbol{\sigma})'\boldsymbol{W}\boldsymbol{g}(\boldsymbol{\sigma})$ with moments $g_m(\boldsymbol{\sigma}) = \sum_j \xi_j(\boldsymbol{\sigma}) z_{m,j}$.
6. Repeat steps 2 to 5 for alternative trials of $\boldsymbol{\sigma}$ until $\boldsymbol{g}(\boldsymbol{\sigma})'\boldsymbol{W}\boldsymbol{g}(\boldsymbol{\sigma})$ is minimized.

Simulation: need for high accuracy in BLP context

- There are two sources of approximation error. The first is driven by the number of simulation draws, while the second is the stopping rule in the NFXP contraction.
- In practice parameter estimates can be sensitive to the stopping rule settings used and to the number of draws in simulation. This is a problem in BLP (rather than GMM with simulated moments more generally) because it combines simulated market shares and the NFXP inversion routine, which propagates the error from one j to another.
- The recent paper Brunner et al (2023) suggests that crude numerical integration of the market share is the main driver of these problems. And they go away with a large enough number of simulation draws. They discuss a number of techniques for taking simulation draws (rather than just Monte Carlo with random draws) that improve the accuracy.

Instruments I

- Consider the error in Mixed, Nested, and Simple Logit models:

$$\xi_j = \begin{cases} \delta_j(\boldsymbol{s}, \boldsymbol{\sigma}) - (\beta \boldsymbol{x}_j - \alpha p_j) & \text{ML} \\ (\log s_j - \log s_0) - \sigma \ln(s_j/s_r) - (\beta \boldsymbol{x}_j - \alpha p_j) & \text{NL} \\ (\log s_j - \log s_0) - (\beta \boldsymbol{x}_j - \alpha p_j) & \text{SL} \end{cases}$$

- Moving up the three expressions, we successively relax IIA property, which implies more market share variables on the RHS.
- Throughout the course we assume

$$E(\xi_{jt}(\boldsymbol{\theta}_0)|\boldsymbol{x}_t, \mathrm{w}_t) = 0,$$

where w_t are cost-shifters. Market shares and prices are not assumed exogenous.

Instruments II

- Exogeneity of x_t
- The $\boldsymbol{x}_j$ are chosen by the firm so why are these assumed exogenous (unlike price)?
- This question is relevant for all specifications of canonical model.
- The argument is that $\boldsymbol{x}_j$ is chosen in advance, before ξ_{jt} observed.
- e.g. for cars: design chosen years in advance.
- And characteristics are time-invariant for any j: $\boldsymbol{x}_t = (x_j)_{j \in J_t}$.
- This is only a partial defense since firms might be forward-looking and could anticipate the realization of ξ_{jt}.
  - This can be dealt with, if we observe panel data, by assuming

$$\xi_{jt} = \rho \xi_{jt-1} + \Delta \xi_{jt} \qquad E[\Delta \xi_{jt} | \boldsymbol{x}_t, \mathrm{w}_t] = 0.$$

  - Another panel option is the fixed-effect specification, where

$$\xi_{jt} = \xi_j + \Delta \xi_{jt} \qquad E[\Delta \xi_{jt} | \boldsymbol{x}_t, \mathrm{w}_t] = 0.$$

Instruments III

- The assumption
$$E(\xi_{jt}(\boldsymbol{\theta}_0)|\boldsymbol{x}_t, \mathrm{w}_t) = 0,$$
implies M zero-covariance conditions
$$E(\xi_{jt}\boldsymbol{z}_{jt}) = 0$$
where
$$\boldsymbol{z}_{jt} = \boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$$
for any instrument function $\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$.
- $\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$ defines M instruments as functions of exogenous data.
- There are many potential choices of $\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$ (with different implications for the number M) which differ in extent of correlation with endogenous variables.
- We now consider some alternatives.

Instruments IV

- BLP instrument function $\boldsymbol{z}_{jt} = \boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$ such that

$$\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t) = \begin{cases} \mathrm{w}_{jt} & \\ \boldsymbol{x}_j^{\text{own}} & = (x_{1j}, \ldots, x_{Kj}) \\ \boldsymbol{x}_{jt}^{\text{samef}} & = (\sum_{\{l: l \neq j, l \in \mathcal{J}_{ft}\}} x_{1l}, \ldots, \sum_{\{l: \neq j, l \in \mathcal{J}_{ft}\}} x_{5l}) \\ \boldsymbol{x}_{jt}^{\text{otherf}} & = (\sum_{\{l: l \notin \mathcal{J}_{ft}\}} x_{1l}, \ldots, \sum_{\{l: l \notin \mathcal{J}_{ft}\}} x_{5l}) \end{cases}$$

  where (i) there are K characteristics and (ii) $\mathcal{J}_{ft}$ is firm $f(j)$'s products in market t.
- Instruments vary across t if there's entry/exit.
- Intuition for this choice of instrument function:
  - w_t shifts price (identifies α).
  - $\boldsymbol{x}_j^{\text{own}}$ are instruments for themselves (identify β).
  - $\boldsymbol{x}_j^{\text{samef}}$ and $\boldsymbol{x}_j^{\text{otherf}}$ shift shares $\boldsymbol{s}$ (and identify $\boldsymbol{\sigma}$).
- If a cost-shifter w_t is absent then $\boldsymbol{x}_{jt}^{\text{samef}}$ and $\boldsymbol{x}_{jt}^{\text{otherf}}$ also shift p_{jt}.
- In an ML setting we no longer motivate BLP instruments as being only for price (as we did in SL).

Instruments V

- Differentiation instruments design $\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t)$ to capture how crowded or isolated product j is in characteristics space.
- Similar to motivation for "nest share" instrument in nested logit.
- These instrument use the vector of characteristic differences between product j and j' in market t:

$$\boldsymbol{d}_{jj't} = \boldsymbol{x}_{jt} - \boldsymbol{x}_{j't}$$

- Rough intuition: if these differences are large, product j is isolated from other products.
- Can be better than BLP instruments; see discussion in Gandhi & Houde (2020) or simpler discussion in Gandhi & Nevo (2021)
- There are different instrument functions for these characteristic differences. We consider one on the next side.

Instruments VI

- One option is based on squares and interactions:

$$\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t) = \begin{cases} \mathrm{w}_{jt} & & \text{Price IVs} \\ \sum_{j' \neq j} (d^k_{jj',t})^2 & \forall k & \text{Isolation in } x_k \\ \sum_{j' \neq j} (d^k_{jj',t} \times d^l_{jj',t}) & \forall k \neq l & \text{Interact: } k \text{ and } l \end{cases}$$

where

1. the price IV is a cost-shifter
2. the isolation in x_k IV is the sum of square differences with other products, a continuous measure of j' isolation along dimension k.
3. interaction terms capture covariance between two dimensions.

Instruments VII

- Another option is based on indicators for whether other products are within a fixed difference κ^k of j:

$$\boldsymbol{A}_j = \begin{cases} \mathrm{w}_{jt} & \text{Price IVs} \\ \sum_{j' \neq j} 1(|d^k_{jj',t}| < \kappa^k) \quad \forall k & \text{Isolation in } x_k \\ \sum_{j' \neq j} 1(|d^k_{jj',t}| < \kappa^k) 1(|d^l_{jj',t}| < \kappa^l)) \quad \forall k \neq l & \text{Interact } k \text{ \& } l \end{cases}$$

where the intuition is similar intuition to before: where

1. the price IV is a cost-shifter
2. the isolation in x_k IV is the number of "neighbour" products in each k (low if j is isolated, high if crowded).
3. interaction terms capture correlation in characteristics for which products are neighbours.

- This approach can be more helpful (than the previous one based on squares) when characteristics are discrete.

Instruments VIII

- An alternative approach to $A_j(\boldsymbol{x}_t, \mathrm{w}_t)$ is optimal instruments .
- As we previously stated, Chamberlain (1987) shows optimal instruments are

$$\boldsymbol{A}_j(\boldsymbol{x}_t, \mathrm{w}_t) = E\left[\frac{\partial \xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t)}{\partial \theta'} | \boldsymbol{x}_t, \mathrm{w}_t\right] \qquad (1 \times L).$$

- This is for single-equation GMM i.e. no supply side equation being estimated, see Reynaert and Verboven (2014, sec 2.5).
- The optimal instruments have the same dimension L as the parameter vector.
- Intuitively, they exploit the functional forms of the model.
- Need unknown θ_0. But, can approximate as follows (see Working Paper 2012 version of Reynaert and Verboven (2014)).
- For simplicity let there be marginal cost pricing: $p_{jt} = \mathrm{w}_{jt}$.

Instruments IX

- Optimal instruments for linear parameters are

$$E\left[\frac{\partial \xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t)}{\partial \alpha}|\boldsymbol{x}_t, \mathrm{w}_t\right] = E\left[p_{jt}|\boldsymbol{x}_t, \mathrm{w}_t\right] = \mathrm{w}_{jt}$$
$$E\left[\frac{\partial \xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t)}{\partial \boldsymbol{\beta}'}|\boldsymbol{x}_t, \mathrm{w}_t\right] = E\left[\boldsymbol{x}_j|\boldsymbol{x}_t, \mathrm{w}_t\right] = x_j$$

- Optimal instruments for nonlinear parameters are

$$E\left[\frac{\partial \xi_{jt}(\boldsymbol{\theta}_0, \boldsymbol{s}_t)}{\partial \boldsymbol{\sigma}'}|\boldsymbol{x}_t, \mathrm{w}_t\right] = E\left[\frac{\partial \delta_{jt}(\boldsymbol{\sigma}_0, \boldsymbol{s}_t)}{\partial \boldsymbol{\sigma}'}|\boldsymbol{x}_t, \mathrm{w}_t\right] \tag{67}$$

  a non-linear function of the $\boldsymbol{x}_j$ of all competing products.

- This expectation is a function of the true $\boldsymbol{\theta}_0$ so cannot be computed directly from the data. Require a first-stage estimate of the demand model.

Instruments X

- BLP (1999) (*AER*) do this by replacing the expected value of the derivatives in (67) by the derivatives evaluated as follows:
  1. Obtain an initial estimate $\hat{\boldsymbol{\theta}}$ (based on BLP or differentiation IV).
  2. Compute the predicted mean utility $\hat{\delta}_j = \boldsymbol{x}_j\hat{\boldsymbol{\beta}} - \hat{\alpha}\mathrm{w}_j$ (i.e. setting the unknown $\xi_j = 0$, its expected value) and predicted market shares $\hat{\boldsymbol{s}}_t$.
  3. Compute the Jacobian of the mean utility evaluated at $\hat{\boldsymbol{\sigma}}$:

$$\left.\frac{\partial \delta_{jt}(\boldsymbol{\sigma}, \hat{\boldsymbol{s}}_t)}{\partial \boldsymbol{\sigma}'}\right|_{\hat{\boldsymbol{\sigma}}}$$

     This can be done using the implicit function theorem by numerically differentiating market share wrt δ and σ respectively.
- More recent literature improves on this approximation by averaging over draws of ξ in the second step.
- Reynaert and Verboven (2014) report that these instruments perform well in simulation exercises.

Figure: Ford Sierra Sapphire

Mixed logit application 1: BLP (1995) on autos I

- Here we outline the specification of BLP's demand model.
- In fact BLP jointly estimated a demand model and a model of price-setting (a supply side). We will for now proceed as in the BLP paper but without the supply side.
- Data as for simple logit model example around slide 72 above, with five product characteristics: $\boldsymbol{x}_j = (1, hpwt, air, size, mpd)$.
- The panel is unbalanced: there are different models j in different years t; this is helpful for identifying parameters that govern substitution patterns, as it allows the data to show how market shares change when choice sets vary.

Mixed logit application 1: BLP (1995) on autos II

- The BLP functional form. We will specify that the utility for consumer i in market t from product j is

$$u_{ijt} = \delta_{jt} + \sum_{k=1}^{5} \sigma_k \nu_{ikt} x_{jk} - \alpha \frac{p_{jt}}{y_{it}} + \varepsilon_{ijt} \tag{68}$$

$$\delta_{jt} = \boldsymbol{\beta} \boldsymbol{x}_j + \xi_{jt} \tag{69}$$

$$y_{it} = \exp(\hat{m}_t + \hat{\sigma}_y \nu_{iyt}) \tag{70}$$

$(\nu_{i1t}, \ldots, \nu_{i5t}, \nu_{iyt})$ are iid standard normal draws, $(\hat{m}_t, \hat{\sigma}_k)$ are pre-estimated mean and spread terms for the empirical distribution of income, and other notation is as previously defined.

- BLP (1995) actually use

$$u_{ijt} = \delta_{jt} + \sum_{k=1}^{5} \sigma_k \nu_{ikt} x_{jk} + \alpha \ln(y_{it} - p_{jt}) + \varepsilon_{ijt}$$

where $\alpha \ln(y_{it} - p_{jt})$ can be interpreted as utility from non-car consumption; this approximates to (68) for small p_{jt}/y_{jt}.

Mixed logit application 1: BLP (1995) on autos III

- The specification of income y_{it} comes from the assumption that household income has a lognormal distribution in the population.
- 11 parameters:
  - 5 linear parameters: β.
  - 6 nonlinear parameters: $\boldsymbol{\sigma} = (\sigma_{i1}, \ldots, \sigma_{iK}, \alpha)$.
- The market share $\boldsymbol{s}^{sim}(\boldsymbol{\delta}, \boldsymbol{\sigma})$ is obtained by Monte Carlo integration, simulating over the normal draws.
- The 15 instruments, z_{jtm} for $m = 1, \ldots, 15$, introduced for the simple logit model (around page 72) are used again, giving 15 covariance sample moments $g(\beta, \boldsymbol{\sigma}) = \sum \xi_{jt} z_{jtm}$.
- The $\boldsymbol{\xi}$ for the moments are obtained by (i) inverting $\boldsymbol{s} = \boldsymbol{s}^{sim}(\boldsymbol{\delta}, \boldsymbol{\sigma})$ to find $\boldsymbol{\delta}$ and (ii) using $\xi_{jt} = \delta_{jt} - \beta \boldsymbol{x}_j$.
- The linear parameters β are concentrated out as there is a linear form for $\beta(\boldsymbol{\sigma})$. Hence, the nonlinear search is for $\boldsymbol{\sigma}$.
- Standard errors are clustered on the 997 products.

Mixed logit application 1: BLP (1995) on autos IV

- Classwork 4 session.
- Estimatng the BLP autos model using demand side only.
- Because the dataset in `BLPestimatoR` does not have income draws we will have to draw these ourselves! We will use the form for utility a couple of slides ago where price interacts the reciprocal of income.
- Tasks include: estimating parameters and standard errors in `BLPestimatoR`. Calculating elasticities and checking for their plausibility.

Figure: Corn Flakes (Kelloggs) and Weetabix (Post Holdings)

Mixed logit application 2: Nevo (2001) on cereals I

- The data in Nevo (2001) has this structure:
  - $J = 25$ brands of breakfast cereal, j.
  - 20 quarters (1988:Q1 - 1992:Q4), $q(t)$.
  - 65 cities, increasing over time, $c(t)$.
  - 1124 markets (city-quarter combinations), t.
- Market shares and prices:
  - Quantity q_{jt} is the total number of 'servings' sold.
  - Market size I_t is inhabitants $\times$ days in a quarter, i.e. one serving per capita per day.
  - Given q_{jt} and I_t, market shares of j follow.
  - Price p_{jt} is total value of sales of the brand at t (deflated by regional urban CPI) divided by q_{jt}. Price varies across cities as well as quarters.
- Vector of product characteristics:
  $x_j =$ [cal from fat, sugar, mushy, fiber, all, children, adult] (where the last three variables are dummies for products geared to different consumer segments) does not vary across t.

Mixed logit application 2: Nevo (2001) on cereals II

- Consumer i in market t utility from product j:

$$u_{ijt} = \delta_{jt} + \sum_{k=1}^{K}(\boldsymbol{\pi}_k \boldsymbol{D}_{it} + \sigma_k \nu_{ikt})x_{jk} - (\boldsymbol{\pi}_0 \boldsymbol{D}_{it} + \sigma_0 \nu_{i0t})p_{jt} + \varepsilon_{ijt}$$
$$\delta_{jt} = d_j + d_{q(t)} - \alpha p_{jt} + \Delta\xi_{jt}$$
$$d_j = \boldsymbol{x}_j \boldsymbol{\beta} + \xi_j$$

  and $u_{i0t} = \varepsilon_{i0t}$ where
  - $\boldsymbol{D}_{it}$ is a vector of i's demographics (income, incomesq, age, and child)
  - $\boldsymbol{\pi}_k$ ($\boldsymbol{\pi}_0$) is a parameter vector
  - d_j and $d_{q(t)}$ are dummy variables for the brand and quarter, $q(t)$
  - Unobserved utility has two components: ξ_j and $\Delta\xi_{jt}$.

  and other notation is as defined before.
- The linear parameters that are concentrated out are $(d_1, ..., d_J, d_{q(1)}, ..., d_{q(19)}, \alpha)$.
- Number of simulation draws $NS = 40$, for demographics $\boldsymbol{D}_{it}$ and N(0,1) shocks $\boldsymbol{\nu}_{it}$.

Mixed logit application 2: Nevo (2001) on cereals III

- Estimation is similar to in BLP, but with four main differences, enumerated (1),...,(4):
- (1) The error term that goes into the moments is $\Delta\xi_{jt}$, i.e. the city-quarter specific deviation after controlling for the average (across city-quarters) unobserved quality ξ_j through the brand fixed effects d_j.
- The moments used for GMM estimation are $z_{jt}\Delta\xi_{jt}$, where

$$\Delta\xi_{jt} = \delta_{jt}(\sigma) - [d_j + d_{q(t)} - \alpha p_{jt}]$$

  and z_{jt} is a vector of instruments.
- This means that Nevo assumes $E[z_{jt}\Delta\xi_{jt}] = 0$, but does not need BLP's assumption $E[z_{jt}\xi_{jt}] = 0$ for his main results.
- This is an advantage: ξ_{jt} is more likely to be correlated with j-specific instruments than the city-quarter deviation $\Delta\xi_{jt}$.

Mixed logit application 2: Nevo (2001) on cereals IV

- (2) Whereas BLP assumed independence across j but not t, Nevo assumes independence across each jt-observation.
- BLP has 999 j-observations, so can rely on asymptotics in the j-dimension. But Nevo has only 25 j-observations, so if he defined the moments like BLP do, there would be only 25 observations of each moment.

Mixed logit application 2: Nevo (2001) on cereals V

- (3) Note that, unlike BLP, β and ξ are not estimated in the main GMM estimation, as they do not enter the model except via brand dummies d_j (which are estimated).
- β and ξ can't be separately identified in the first step because product characteristics do not vary across city-quarters (t).
- To break out d_j into the effects of x_j and ξ_j a second stage of estimation is needed, where d_j is regressed on x_j and the residuals are ξ_j, i.e.

$$d_j = x_j\beta + \xi_j.$$

  This is estimated linearly using the assumption $E(x_j\xi_j) = 0$.

Mixed logit application 2: Nevo (2001) on cereals VI

- (4) Instruments differ from BLP's. The instrument vector z_{jt} contain brand and quarter dummy variables which help to identify brand and quarter dummy parameters.
- The remaining parameters (for which other instruments are needed) are
  - α (1 parameter)
  - σ (9 parameters)
  - π (10 parameters).
- Product characteristics do not vary across city-quarters (t), so are co-linear with the brand dummy variables. Therefore, the BLP instruments cannot be used alongside brand dummies as instruments.

Mixed logit application 2: Nevo (2001) on cereals VII

- Instead, Nevo uses the following instruments, for each (j, t):
  - Hausman instruments: The quarterly average price of j in other cities in the same region as the city in the city-quarter t. Since there are 20 quarters in the data, and each quarterly average is included as a separate instrument this gives 20 instruments. (Footnote 21 in his paper: "The results are essentially the same if I use only the regional average price for that quarter".)
  - Proxies for marginal cost: census region dummy variables, city density and average city earnings in the supermarket sector. This gives 10 additional moment conditions.
- Both sets of instruments should be helpful to estimate the price parameter. The paper does not discuss how they contribute to estimating the σ and π parameters, but seems plausible that instruments that shift price and therefore market shares will provide information, since σ and π determine substitution patterns.

Mixed logit application 2: Nevo (2001) on cereals VIII

- Homework exercise.
- Because of time constraints we will not do work in class time on the Nevo model.
- However, `BLPestimatoR` contains fake breakfast cereal data that has been generated by Nevo that is similar in structure to the data in his paper.
- The data includes draws of demographics.
- As (optional) homework, it is worth in your own time following the R steps that are detailed in the BLPestimatoR vignette to replicate the estimates they present and calculate elasticities: see in classwork5_cereals.pdf on course webpage.

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Multi-product Nash pricing and implied marginal costs I

- Often we are interested in analyzing the first order conditions implied by the estimated demand system.
- This may be to measure market power (given by the markups at the observed market structure) or to simulate a counterfactual equilibrium from a policy change.
- For a multi-product firm f with ownership J_f the profit is

$$\Pi_f = \sum_{j \in J_f} s_j \times (p_j - mc_j)$$

- The first order condition for product j is

$$s_j + \sum_{l \in J_{f(j)}} (p_l - mc_l) \frac{\partial s_l}{\partial p_j} = 0 \tag{71}$$

or

$$s_j + \sum_{l \in J} \mathcal{H}_{jl} (p_l - mc_l) \frac{\partial s_l}{\partial p_j} = 0 \tag{72}$$

where $\mathcal{H}_{jl} = 1$ if (j, l) are co-owned and $\mathcal{H}_{jl} = 0$ otherwise.

Multi-product Nash pricing and implied marginal costs II

- Let $\boldsymbol{\Delta}(\mathcal{H})$ be $J \times J$ such that term in jth row and lth column is

$$\Delta_{jl}(\mathcal{H}) = \mathcal{H}_{jl} \frac{\partial s_j}{\partial p_l} \tag{73}$$

- Then (72) in vector form is

$$\boldsymbol{s} - \boldsymbol{\Delta}(\mathcal{H})[\boldsymbol{p} - \boldsymbol{mc}] = \mathbf{0}$$

- Then we get the expression for markups:

$$\boldsymbol{p} - \boldsymbol{mc} = [\boldsymbol{\Delta}(\mathcal{H})]^{-1}\boldsymbol{s}. \tag{74}$$

Counterfactual simulation I

- Let us consider a counterfactual ownership matrix $\mathcal{H}^*$.
- Since $\boldsymbol{\Delta}$ depends on prices and ownership we write $\boldsymbol{\Delta}(\mathcal{H}^*, \boldsymbol{p})$.
- First, we back out marginal costs at the observed market structure, using the $\boldsymbol{\Delta}$ implied by the estimated parameters, and the observed prices and market shares:

$$\boldsymbol{mc} = \boldsymbol{p} - [\boldsymbol{\Delta}(\mathcal{H}, \boldsymbol{p})]^{-1}\boldsymbol{s}.$$

- Counterfactual prices $\boldsymbol{p}^*$ solve the following system

$$\boldsymbol{p}^* = \boldsymbol{mc} + [\boldsymbol{\Delta}(\mathcal{H}^*, \boldsymbol{p}^*)]^{-1}\boldsymbol{s}(\boldsymbol{p}^*).$$

- This system is nonlinear and is solved by numerical methods. There are three alternatives: (i) use an equation solver, (ii) iterate the system to convergence, (iii) use the reformulation of the problem in Morrow and Skerlos (2011) *Operations Research*.

Counterfactual simulation II

- In practice (i) sometimes fails, (ii) may fail as it is not a contraction, and (iii) appears reliable (as it is a contraction).
- Finally, there is no result showing the existence or uniqueness of prices implied by the first-order conditions. Usually authors assume existence and uniqueness. In practice this does not seem to matter: it is rare not to find an equilibrium.

Classwork 6

The fourth exercise simulates counterfactual prices for a merger of Chrysler and GM in the auto industry. See classwork6_merger.pdf on course webpage.

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Estimating a supply side I

- We now consider joint estimation of the demand and pricing equations.
- This is done in the main specification in BLP, and we will use their example to make the discussion concrete.
- Note that the estimation of a pricing equation can help identify demand as well as marginal cost.
- This benefit comes at the cost of having to make an assumption about how prices are set. The standard assumption is multi-product Nash.

Estimating a supply side II

- Suppress market subscripts t. Let marginal cost of j be

$$mc_j = \boldsymbol{\gamma} \boldsymbol{x}_j^c + \omega_j$$

where $\boldsymbol{x}_j^c$ is a vector of observable cost-shifters including a constant term and ω_j is an unobserved cost scalar.

- Let the expression for price-cost markups in 74 be denoted η:

$$\boldsymbol{\eta}(\boldsymbol{\sigma}, \boldsymbol{\delta}, \alpha) = [\boldsymbol{\Delta}(\mathcal{H})]^{-1} \boldsymbol{s}. \tag{75}$$

where we write markup η to emphasise: (i) it's a function of demand parameters, (ii) the $\boldsymbol{\beta}$ parameters on non-price characteristics enter only via $\boldsymbol{\delta}$, and (iii) the price parameter α now enters separately from δ (via the derivative wrt price).

- For product j

$$p_j - \eta_j = \boldsymbol{\gamma} \boldsymbol{x}_j^c + \omega_j \tag{76}$$

Estimating a supply side III

- We assume that there is an $M_c \times 1$ vector $\boldsymbol{z}_m^c$ such that in the population $E(\omega|\boldsymbol{z}^c) = 0$ and $E(\omega) = 0$, which implies

$$E[z_m^c \omega] = 0 \text{ for } m = 1, \ldots, M_c. \tag{77}$$

- The corresponding sample moment conditions are

$$g_m^c(\boldsymbol{\theta}, \gamma) = \sum_{j=1}^{J} \omega_j z_{mj}^c \text{ for } m = 1, \ldots, M_c$$

- We stack the demand and supply side moments, i.e.

$$\boldsymbol{g}(\boldsymbol{\theta}, \gamma) = \begin{bmatrix} g_1(\boldsymbol{\theta}) \\ \vdots \\ g_M(\boldsymbol{\theta}) \\ g_1^c(\boldsymbol{\theta}, \gamma) \\ \vdots \\ g_{M_c}^c(\boldsymbol{\theta}, \gamma) \end{bmatrix} \tag{78}$$

Estimating a supply side IV

and use

$$(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\gamma}}) = \arg\min_{(\boldsymbol{\theta},\boldsymbol{\gamma})} \boldsymbol{g}(\boldsymbol{\theta}, \boldsymbol{\gamma})' \boldsymbol{W} \boldsymbol{g}(\boldsymbol{\theta}, \boldsymbol{\gamma}) \tag{79}$$

where there are now $M + M_c$ moments.

- Concentrating out linear parameters. As equation (76) is linear in γ parameters, and resembles a linear regression form, γ be concentrated out conditional on nonlinear parameters $(\boldsymbol{\sigma}, \alpha)$ using the linear IV estimators.
- The price parameter α is now nonlinear because it enters the markup term non-linearly.
- There are now therefore two sets of linear parameters to concentrate out: (i) the (non-price) β parameters in demand and (ii) the γ parameters in cost.
- Therefore the nonlinear estimator is:

$$(\hat{\boldsymbol{\sigma}}, \hat{\boldsymbol{\alpha}}) = \arg\min_{(\boldsymbol{\sigma},\alpha)} \boldsymbol{g}(\boldsymbol{\sigma}, \alpha)' \boldsymbol{W} \boldsymbol{g}(\boldsymbol{\sigma}, \alpha) \tag{80}$$

Estimating a supply side V

- What instruments should be used? In the application BLP(1995) the 18 "cost side" instruments are:

$$\boldsymbol{z}_j = (\boldsymbol{x}_j^c, \boldsymbol{x}_j^{c,\text{samef}}, \boldsymbol{x}_j^{c,\text{otherf}}) \tag{81}$$

  where

$$\begin{aligned}\boldsymbol{x}_j^{c,\text{samef}} &= (\textstyle\sum_{\{k:k\neq j, k\in\mathcal{J}_f\}} x_{6j}^c, \ldots, \sum_{\{k:k\neq j, k\in\mathcal{J}_f\}} x_{6j}^c)\\ \boldsymbol{x}_j^{c,\text{otherf}} &= (\textstyle\sum_{\{k:k\notin\mathcal{J}_f\}} x_{1j}^c, \ldots, \sum_{\{k:k\notin\mathcal{J}_f\}} x_{6j}^c)\end{aligned}$$

  where $\mathcal{J}_f$ is products produced by firm (f) that produces j.
- BLP had 6 variables in $\boldsymbol{x}_j^c$: a constant, $log(hpwt)$, Air, $log(MPG)$, $log(size)$, and a trend term, giving 18 instruments interacted with ω.
- Since price enters the supply side too, it makes sense to use instruments that co-vary with it, so the same market structure / product ownership rationale applies here as for the demand side.
- Overall there are 15 instruments interacted with ξ and 18 interacted with ω to estimate 17 model parameters.

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Adding consumer-level data I

- Ref: Train (2009), BLP (2004).
- Suppose in addition to the market level data

$$\{\boldsymbol{z}_j, s_j, \boldsymbol{x}_j, p_j\} \text{ for } j \in \{0, 1, ..., J\}$$

we have the following random sample of consumers $i = 1, \ldots N$ with variables

$$\{\boldsymbol{h}_i\} \text{ for } i = 1, ..., N \text{ [e.g. household size, location,...]}$$
$$\{y_{ij}\} \text{ for } i = 1, ..., N \text{ and } j \in \{0, 1, ..., J\}$$

where

$$y_{ij} = \begin{cases} 1 & \text{if } i \text{ chooses } j \\ 0 & \text{otherwise.} \end{cases}$$

Adding consumer-level data II

- Let preferences be given by

$$u_{ij} = \sum_{k=1}^{K}(\beta_k + \boldsymbol{\sigma}_k \boldsymbol{h}_i)x_{jk} - (\alpha + \boldsymbol{\sigma}_0 \boldsymbol{h}_i)p_j + \xi_j + \varepsilon_{ij} \tag{82}$$

$$\delta_j = \boldsymbol{\beta}\boldsymbol{x}_j - \alpha p_j + \xi_j \tag{83}$$

where $\varepsilon \sim G(\varepsilon; \boldsymbol{\sigma}_\varepsilon)$ so that probability "i choose j" is

$$P_{ij} = \int_{\varepsilon} 1(u_{ij} \geq u_{ik} \forall k \neq j) dG.$$

- Let $(z_{ij,1}, \ldots, z_{ij,M_a})$ be M_a instruments defined at (i, j) level.
- Assume that the following moment conditions hold:

$$g_m^a = \sum_i \sum_j [y_{ij} - P_{ij}] z_{ij,m} = 0 \text{ for } m = 1, \ldots, M_a \tag{84}$$

where the a superscript distinguishes the micro moments from other moments.

Adding consumer-level data III

- Example (auto choice): let the instrument in the first moment ($m = 1$) be $z_{ij,1} = hz_i \times space_j$ where hz_i is household size, $space_j$ is space in the car.
- The moment condition for this instrument matches the observed and predicted covariance between i's household size and space of i's chosen car, as the following steps illustrate:

$$\begin{aligned} g_1^a &= \sum_i \sum_j [y_{ij} - P_{ij}] hz_i space_j \\ &= \sum_i hz_i \sum_j [y_{ij} space_j - P_{ij} space_j] \\ &= \sum_i hz_i space_i^{obs} - \sum_i hz_i space_i^{pred} \\ &= 0 \end{aligned}$$

where $space_i^{obs} = \sum_j y_{ij} space_j$ is the size of car observed to be chosen by i and $space_i^{pred} = \sum_j P_{ij} space_j$ is the expected size of car chosen by i as predicted by the model.

Adding consumer-level data IV

- Such moments are informative about the parameter on the interaction between household attributes and product characteristics.
- Other instruments can employ observed
  - distance from household to choice (relevant for store or school choices for example)
  - ranked choice data on the first-best choice and runner-up choice of the household (see BLP (2004)).

Adding consumer-level data V

- We stack the micro moments $g_1^a, \ldots, g_{M_a}^a$ with the demand and supply side moments, i.e.

$$\boldsymbol{g}(\boldsymbol{\sigma}, \alpha) = \begin{bmatrix} g_1(\boldsymbol{\sigma}, \alpha) \\ \vdots \\ g_M(\boldsymbol{\sigma}, \alpha) \\ g_1^c(\boldsymbol{\sigma}, \alpha) \\ \vdots \\ g_{M_c}^c(\boldsymbol{\sigma}, \alpha) \\ g_1^a(\boldsymbol{\sigma}, \alpha) \\ \vdots \\ g_{M_a}^a(\boldsymbol{\sigma}, \alpha) \end{bmatrix} \tag{85}$$

and proceed as before, i.e.

$$(\hat{\boldsymbol{\sigma}}, \hat{\boldsymbol{\alpha}}) = \arg\min_{(\boldsymbol{\sigma}, \alpha)} \boldsymbol{g}(\boldsymbol{\sigma}, \alpha)' \boldsymbol{W} \boldsymbol{g}(\boldsymbol{\sigma}, \alpha). \tag{86}$$

Additional instruments to interact with ξ may be available with micro data

- Additional instruments to interact with ξ may be made available when consumer-level data is used: e.g. so-called Waldfogel instruments. These are market-level averages of consumer attributes. For example income: prices in high-income markets may be unusually high. Such instruments may be excludable with micro data if the choice probabilities in the model condition on the consumer's own observed value of the attribute e.g. income.

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Nonparametric Identification

Nonparametric Identification with Market-Level Data I

- Motivating question: are the σ parameters for random coefficients in mixed logit models identified?
- We now discuss Berry-Haile's formal argument showing all parameters are identified nonparametrically with market level data.
  - requires one of the x not to have a random coefficient (a "special regressor")
  - does not mean we want to *estimate* non-parametrically but it does reassure us that we are not relying on functional forms to identify parameters
- Suppose we have market level data for markets $t = 1, .., T$:

$$\{\boldsymbol{z}_{jt}, s_{jt}, \boldsymbol{x}_{jt}, p_{jt}\} \text{ for } j \in \{0, \ldots, J\}$$

  - We suppress the market subscript t.

Nonparametric Identification with Market-Level Data II

- Consider the canonical model:

$$u_{ij} = x_j + \boldsymbol{\beta}_i \boldsymbol{x}_j^{(2)} - \alpha_i p_j + \xi_j + \varepsilon_{ij} \tag{87}$$

$$\delta_j = x_j + \xi_j \tag{88}$$

  where

  - α and β are heterogeneous and we now leave unspecified the parametric form of the variation
  - $\boldsymbol{x}_j = (x_j, \boldsymbol{x}_j^{(2)})$
  - $E[\xi_j | \boldsymbol{x}_j] = 0$ (as BLP)
- Note x_j has no random coefficient (special regressor assumption)
- Note also the following innocuous changes to notation:
  - utility is scaled so the parameter on x_j is 1
  - index now defined: $\delta_j = x_j + \xi_j$—i.e. the $x^{(2)}$ are now not included in the index.
- From now (for simplicity) we condition on $x^{(2)}$ and drop it from notation

Nonparametric Identification with Market-Level Data III

- The argument is in 3 steps:

  Index Define index $\delta_j = x_j + \xi_j$ where x_j is a scalar that enters u_{ij} *only* via δ_j.

  Invert Invert $s_j = s_j(\boldsymbol{\delta}, \boldsymbol{p})$ and rearrange to get x_j on left-hand-side and ξ_j on right: $x_j = \delta_j(\boldsymbol{s}, \boldsymbol{p}) - \xi_j$.

  Instrument Interpret expression $x_j = \delta_j(\boldsymbol{s}, \boldsymbol{p}) - \xi_j$. in step 2 as a nonlinear regression with a separable error and apply existing nonparametric identification arguments for this case (Newey and Powell (2003)).

- We now see why this requires a special regressor: x_j is treated as a 'LHS variable' in a regression whose (market level) unit of observation is the product j.
- We discuss each step in more detail...

Nonparametric Identification with Market-Level Data IV

Step 1: Index.

- Utility is:

$$u_{ij} = \beta x_j - \alpha_i p_j + \xi_j + \varepsilon_{ij}. \tag{89}$$

Normalize the coefficient on x_j to 1 so

$$u_{ij} = x_j - \alpha_i p_j + \xi_j + \varepsilon_{ij} \tag{90}$$

where we implicitly rescale the other parameters.

- Index is:

$$\delta_j = x_j + \xi_j \tag{91}$$

where exclude p_j and characteristics other than x_j from δ_j.

- The market share functions to be identified can be written:

$$\boldsymbol{s} = \boldsymbol{s}(\boldsymbol{\delta}, \boldsymbol{p}). \tag{92}$$

Nonparametric Identification with Market-Level Data V

Step 2: Invert to obtain a Regression Form

- Invert this market share function (92) to isolate $\boldsymbol{\delta}$

$$\boldsymbol{\delta} = \boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{p})$$

- Using $\delta_j = x_j + \xi_j$ from (91) rewrite this (for product j):

$$x_j = \delta_j(\boldsymbol{s}, \boldsymbol{p}) - \xi_j. \tag{93}$$

- This equation
  - has an additive separable error: one ξ_j per observation j
  - two sets of J endogenous variables: $\boldsymbol{s}$ and $\boldsymbol{p}$ per observation j
  - special regressor x_j is excluded from the function $\delta_j(\boldsymbol{s}, \boldsymbol{p})$
  - takes the same form as regression equations for which identification results have been derived by Newey and Powell (2003).

Nonparametric Identification with Market-Level Data VI

Step 3: Instrument

- Newey and Powell (2003) prove that the separable nonlinear model

$$y = f(\boldsymbol{x}) + \xi \qquad \text{where } E[\xi|\boldsymbol{z}] = 0 \tag{94}$$

  is identified provided instruments $\boldsymbol{z}$ move the endogenous variables $\boldsymbol{x}$ "enough".
- "Enough": any two functions of $\boldsymbol{x}$ can be distinguished from one another through variation in $\boldsymbol{z}$. Similar to the rank condition for identification, this is a condition about the variation of $\boldsymbol{z}$ and $\boldsymbol{x}$.
  - (More formal way of saying this is that $\boldsymbol{z}$ is "complete" with respect to x, i.e. if $E(B(\boldsymbol{x})|\boldsymbol{z}) = 0$ then $B(\boldsymbol{x}) = 0$. Loosely, this implies that no non-trivial function of $\boldsymbol{x}$ has the same expectation for different values of $\boldsymbol{z}$).

Nonparametric Identification with Market-Level Data VII

- The expression Berry-Haile derive by inversion, (93),

$$x_j = \delta_j(\boldsymbol{s}, \boldsymbol{p}) - \xi_j \qquad \text{where } E[\xi_j | \boldsymbol{x}, \boldsymbol{z}] = 0. \tag{95}$$

  resembles the nonlinear model with separable error (94).
- The difference is that the LHS variable x_j is exogenous ($E[\xi_j|\boldsymbol{x}] = 0$).
- Berry-Haile show that Newey-Powell (2003)'s identification result still applies and the same "completeness" condition is needed:
  - exogenous $(\boldsymbol{x}, \boldsymbol{z})$ must move endogenous $(\boldsymbol{s}, \boldsymbol{p})$ "enough": any two functions of $(\boldsymbol{s}, \boldsymbol{p})$ can be distinguished from each other through the variation in $(\boldsymbol{x}, \boldsymbol{z})$.
  - require instruments for $\boldsymbol{s}$ as well as $\boldsymbol{p}$.

Parametric Examples & Discussion

Insights from parametric examples

- We are used to the idea we need instruments for price but Berry and Haile's finding that we also (in general) need instruments for $\boldsymbol{s}$ is new.
- To give intuition as to why we need instruments for $\boldsymbol{s}$ as well as price we consider familiar parametric examples: logit, nested logit, and mixed logit.
- We will find that the more general the substitution permitted in the model, the more market share instruments we need, up to all J in the most general case.
- In each case we follow the three (Berry-Haile) steps
  1. Index: specifying an index $\delta_j = x_j + \xi_j$
  2. Invert: inverting market share to get: $x_j = \boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{p}) - \xi_j$,
  3. Instrument: consider instruments needed for $x_j = \boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{p}) - \xi_j$.

Instruments needed for Simple logit model

- Given utility $u_{ij} = \beta x_j + \alpha p_j + \xi_j + \varepsilon_{ij}$ let $\bar{\alpha} = \alpha/\beta$ and $\bar{\xi}_j = \xi_j/\beta$.
- Step 1: let $\bar{\delta}_j = x_j + \bar{\xi}_j$. (Note this is defined differently to mean utility $\delta_j = \beta x_j - \alpha p_j + \xi_j$).
- Step 2: invert simple logit share function (see earlier slides):

$$\ln s_j - \ln s_0 = \beta x_j - \alpha p_j + \xi_j$$

  which can be rewritten

$$x_j = \frac{1}{\beta}(\ln s_j - \ln s_0) + \bar{\alpha} p_j - \bar{\xi}_j$$

  to resemble a linear regression with error $\bar{\xi}_j$
- Step 3: Two instruments are needed: one for $\ln s_j - \ln s_0$ (we can use x_j) and one for p_j. We can identify assuming $E[\bar{\xi}_j | x_j] = 0$ and $E[\bar{\xi}_j | z_j] = 0$ where z_j is a price instrument.

Instruments needed for Nested logit

- Let $\bar{\alpha} = \alpha/\beta$, $\bar{\sigma}_g = \sigma_g/\beta$ and $\bar{\xi}_j = \xi_j/\beta$ in nested logit model from earlier slides.
- Step 1: let $\bar{\delta}_j = x_j + \bar{\xi}$. (Note this is defined differently to mean utility $\delta_j = \beta x_j - \alpha p_j + \xi_j$).
- Step 2: invert nested logit share (see earlier slides) to get:

$$\ln s_j - \ln s_0 = \beta x_j - \alpha p_j + \sigma_g \ln s_{j|g} + \xi_j$$

  which can be rewritten

$$x_j = \frac{1}{\beta}(\ln s_j - \ln s_0) - \bar{\sigma}_g \ln s_{j|g} + \bar{\alpha} p_j - \bar{\xi}_j.$$

- Step 3: We now need 3 instruments: one for p_j, one for $\ln s_j - \ln s_0$, and (now) one for $s_{j|g}$.
- A model with more general substitution patterns than simple logit requires an extra market share instrument.

Instruments needed for Mixed logit or probit

- Let $\bar{\alpha} = \alpha/\beta$, $\bar{\varepsilon}_{ij} = \varepsilon_{ij}/\beta$ and $\bar{\xi}_j = \xi_j/\beta$ etc.
- Step 1: let

$$u_{ij} = x_j + \bar{\xi}_j - (\bar{\alpha} + \bar{\sigma}\nu_i)p_j + \bar{\varepsilon}_{ij} \tag{96}$$

  and $\bar{\delta}_j = x_j + \bar{\xi}_j$. Market share function is $\boldsymbol{s} = \boldsymbol{s}(\bar{\boldsymbol{\delta}}, \boldsymbol{p})$.
- Step 2: Invert (for product j)

$$x_j = \bar{\delta}_j(\boldsymbol{s}, \boldsymbol{p}) - \bar{\xi}_j. \tag{97}$$

- The vectors $\boldsymbol{s}$ and $\boldsymbol{p}$ are endogenous.
- Step 3: Need instruments for the vectors $\boldsymbol{s}$ and $\boldsymbol{p}$. We can use $\boldsymbol{x}$ as the instrument for $\boldsymbol{s}$.

Intuition for the Berry-Haile result? I

- We always understood that for a demand system one needs instruments for prices $\boldsymbol{p}$.
- What we learn is that we *also* need instruments for $\boldsymbol{s}$.
- Quote from BH: "Formally, this requirement reflects the fact that, in a multigood setting, the quantity demanded of a given good depends on the endogenous prices of all goods, as well as the demand shocks associated with all goods.
- Contrast this with a classical supply and demand model, in which the quantity demanded depends on only one endogenous price and one demand shock."
- The more general discrete choice models allows demand to depend on more of those those demand shocks. Only if IIA holds, as in simple logit, does the ratio of a goods market share to the outside good not depend on these shocks.

Intuition for the Berry-Haile result? II

- For more intuition suppose prices are exogenous (e.g. set randomly). If so that's not enough to identify the demand model

$$x_j = \delta_j(\boldsymbol{s}, \boldsymbol{p}, \boldsymbol{\sigma}) - \xi_j \tag{98}$$

  because $\boldsymbol{s}$ is endogenous. Cost-shifters (classic price instruments) will not serve as instruments for $\boldsymbol{s}$ as they only enter model via prices.
- As BH point out there is by construction a suitable instrument for s_j: x_j which is excluded from the function $\boldsymbol{\delta}(\boldsymbol{s}, \boldsymbol{p})$ we seek to identify.
- BH's result is a positive one showing that the model is identified even using market share level data (i.e. we don't need micro data). However it does require $2J$ instruments.

Non-parametric identification: Micro-Level Data

- Berry-Haile extend their result to micro data and show it helps, by reducing number of instruments needed.
- Let data be: random sample of consumers $i = 1, \ldots N$ in markets $t = 1, \ldots, T$:

$$\begin{aligned} \{\boldsymbol{x}_{jt}, p_{jt}\} & \quad \text{for } j \in \{0, 1, ..., J\} \\ \{\boldsymbol{z}_{it} = (z_{i1t}, \ldots, z_{iJt})\} & \quad \text{for } i = 1, ..., N \text{ [characteristic for } ij] \\ \{y_{ij}\} & \quad \text{for } i = 1, ..., N \text{ and } j \in \{0, 1, ..., J\} \end{aligned}$$

where

$$y_{ij} = \begin{cases} 1 & \text{if } i \text{ chooses } j \\ 0 & \text{otherwise.} \end{cases}$$

- We can think of z_{ijt} as (negative or inverse of) distance between consumer i and product j in market t.

Identification w/ Micro-Level Data: Index

- Let utility be

$$u_{ijt} = (z_{ijt} + \xi_{jt}) - \alpha_i p_{jt} + \varepsilon_{ijt} \tag{99}$$

and let z_{ijt} enter utility via the consumer-specific index

$$\delta_{ijt} = z_{ijt} + \xi_{jt} \tag{100}$$

where we normalize parameter on z_{ijt} to 1 (other terms adjust) and suppress $\boldsymbol{x}_j$ from the function for notation simplicity.

  - within a market $(\boldsymbol{p}_t, \boldsymbol{\xi}_t)$, are fixed but $\boldsymbol{z}_{it}$ varies over i.
  - note that z_{ijt} is still a special regressor: there is no heterogeneity in its coefficient.

- Consumer-specific choice probability and inverse:

$$\boldsymbol{s}_{it} = \boldsymbol{s}(\boldsymbol{\delta}_{it}, \boldsymbol{p}_t) \tag{101}$$

$$\boldsymbol{\delta}_{it} = \boldsymbol{\delta}(\boldsymbol{s}_{it}, \boldsymbol{p}_t) \tag{102}$$

Identification w/ Micro-Level Data: fixing s

- Pick a probability vector denoted $\bar{\boldsymbol{s}}$ and fix on this common share for all markets $t = 1, \ldots, T$.
- Assume that using micro data we observe reduced form choice probability function $\boldsymbol{s}_t(\boldsymbol{z})\ \forall \boldsymbol{z}, t$
  - ...and can use this to get $\bar{\boldsymbol{z}}_t(\bar{\boldsymbol{s}})$—i.e. the $\boldsymbol{z}$ that generates $\bar{\boldsymbol{s}}$ in each market t.
- We fix the common $\bar{\boldsymbol{s}}$ and its implied $\bar{\boldsymbol{z}}_t$ in each t;
- this is equivalent to picking a single consumer type $\bar{\boldsymbol{z}}_t$ in each market, namely the one that has the common choice probability (the selected type will vary across markets).
  - Since this is the case we suppress i subscripts and use $\bar{\boldsymbol{z}}_t, \bar{\boldsymbol{\delta}}_t, \ldots$ etc., to indicate the single selected value for each market.

The identification argument

- Index for selected type $\bar{z}_{jt}$ is: $\bar{\delta}_{jt} = \bar{z}_{jt} + \xi_{jt}$.
- Invert market share function $\boldsymbol{s}(\boldsymbol{\delta}_{it}, \boldsymbol{p}_t)$ at the common share $\bar{\boldsymbol{s}}$ to get $\bar{\delta}_{jt} = \delta_j(\bar{\boldsymbol{s}}, \boldsymbol{p}_t)$ for each j and t.
- We get a nonlinear equation with separable error:

$$\bar{z}_{jt} = \delta_j(\bar{\boldsymbol{s}}, \boldsymbol{p}_t) - \xi_{jt}. \tag{103}$$

- This has same form as a nonlinear regression where $\bar{z}_{jt}$ plays the role of the left hand side variable.
- Since $\bar{\boldsymbol{s}}$ is now fixed across t it is not endogenous and we do not need to instrument for it. We now only need instruments for $\boldsymbol{p}$ in regression specification (103).

Discussion of identification w/ micro data

- Argument on last side relies on the same three components used in the market-level data: index, invert, instrument.
- But now
  - the index uses consumer-product attribute z_{ijt}
  - inversion is done at a choice probability $\bar{\boldsymbol{s}}$ we fix
  - we can observe the $\bar{\boldsymbol{z}}_t$ associated with $\bar{\boldsymbol{s}}$ from micro data in each market
- This tells us micro data has identification benefits: removes need for market share instruments.
- Note however that there is still the need for z to be a special regressor (i.e. it cannot have a heterogeneous coefficient).